12 Analysis of Variance

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▲ **ONE VARIABLE THAT GOOGLE** uses to rank pages on the Internet is page speed, the time it takes for a web page to load into your browser. A source for women's clothing is redesigning their page to improve the images that show its products and to reduce its load time. The new page is clearly faster, but initial tests indicate there is more variation in the time to load. A sample of 16 different load times showed that the standard deviation of the load time was 22 hundredths of a second for the new page and 12 hundredths of a second for the current page. At the .05 significance level, can we conclude that there is more variation in the load time of the new page? (See Exercise 24 and **LO12-1**.)

LEARNING OBJECTIVES

When you have completed this chapter, you will be able to:

LO12-1 Apply the *F* distribution to test a hypothesis that two population variances are equal.

LO12-2 Use ANOVA to test a hypothesis that three or more population means are equal.

LO12-3 Use confidence intervals to test and interpret differences between pairs of population means.

LO12-4 Use a blocking variable in a two-way ANOVA to test a hypothesis that three or more population means are equal.

LO12-5 Perform a two-way ANOVA with interaction and describe the results.

INTRODUCTION

In this chapter, we continue our discussion of hypothesis testing. Recall that in Chapters 10 and 11 we examined the general theory of hypothesis testing. We described the case where a sample was selected from the population. We used the *z* distribution (the standard normal distribution) or the *t* distribution to determine whether it was reasonable to conclude that the population mean was equal to a specified value. We tested whether two population means are the same. In this chapter, we expand our idea of hypothesis tests. We describe a test for variances and then a test that simultaneously compares several population means to determine if they are equal.

LO12-1
Apply the *F* distribution to test a hypothesis that two population variances are equal.

COMPARING TWO POPULATION VARIANCES

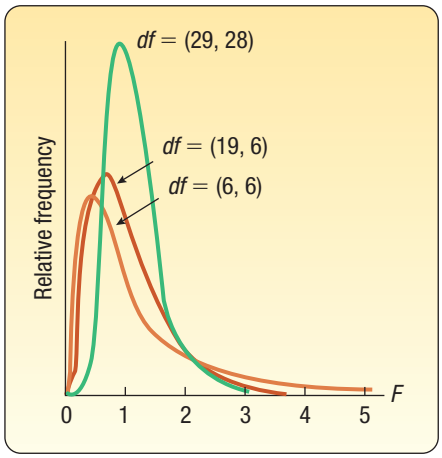
In Chapter 11, we tested hypotheses about equal population means. The tests differed based on our assumptions regarding whether the population standard deviations or variances were equal or unequal. In this chapter, the assumption about equal population variances is also important. In this section, we present a way to statistically test this assumption. The test is based on the *F* distribution.

The *F* Distribution

The probability distribution used in this chapter is the *F* distribution. It was named to honor Sir Ronald Fisher, one of the founders of modern-day statistics. The test statistic for several situations follows this probability distribution. It is used to test whether two samples are from populations having equal variances, and it is also applied when we want to compare several population means simultaneously. The simultaneous comparison of several population means is called **analysis of variance (ANOVA).** In both of these situations, the populations must follow a normal distribution, and the data must be at least interval-scale.

What are the characteristics of the *F* distribution?

1. **There is a family of *F* distributions.** A particular member of the family is determined by two parameters: the degrees of freedom in the numerator and the degrees of freedom in the denominator. The shape of the distribution is illustrated by the following graph. There is one *F* distribution for the combination of 29 degrees of freedom in the numerator (*df*) and 28 degrees of freedom in the denominator. There is another *F* distribution for 19 degrees of freedom in the numerator and 6 degrees of freedom in the denominator. The final distribution shown has 6 degrees of freedom in the numerator and 6 degrees of freedom in the denominator. We will describe the concept of degrees of freedom later in the chapter. Note that the shapes of the distributions change as the degrees of freedom change.

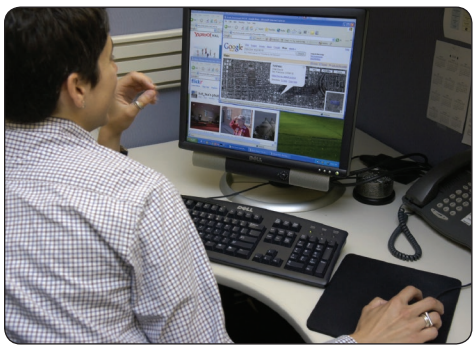


2. **The *F* distribution is continuous.** This means that the value of *F* can assume an infinite number of values between zero and positive infinity.
3. **The *F* statistic cannot be negative.** The smallest value *F* can assume is 0.
4. **The *F* distribution is positively skewed.** The long tail of the distribution is to the right-hand side. As the number of degrees of freedom increases in both the numerator and denominator, the distribution approaches a normal distribution.
5. **The *F* distribution is asymptotic.** As the values of *F* increase, the distribution approaches the horizontal axis but never touches it. This is similar to the behavior of the normal probability distribution, described in Chapter 7.

Testing a Hypothesis of Equal Population Variances

The first application of the *F* distribution that we describe occurs when we test the hypothesis that the variance of one normal population equals the variance of another normal population. The following examples will show the use of the test:

- A health services corporation manages two hospitals in Knoxville, Tennessee: St. Mary's North and St. Mary's South. In each hospital, the mean waiting time in the Emergency Department is 42 minutes. The hospital administrator believes that the St. Mary's North Emergency Department has more variation in waiting time than St. Mary's South.



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- The mean rate of return on two types of common stock may be the same, but there may be more variation in the rate of return in one than the other. A sample of 10 technology and 10 utility stocks shows the same mean rate of return, but there is likely more variation in the technology stocks.
- An on-line newspaper found that men and women spend about the same amount of time per day accessing news apps. However, the same report indicated the times of men had nearly twice as much variation compared to the times of women.

The *F* distribution is also used to test the assumption that the variances of two normal populations are equal. Recall that in the previous chapter the *t* test to investigate whether the means of two independent populations differed assumes that the variances of the two normal populations are the same. See this list of assumptions on page 361. The *F* distribution is used to test the assumption that the variances are equal.

To compare two population variances, we first state the null hypothesis. The null hypothesis is that the variance of one normal population, σ_1^2, equals the variance of another normal population, σ_2^2. The alternate hypothesis is that the variances differ. In this instance, the null hypothesis and the alternate hypothesis are:

$$H_0: \sigma_1^2 = \sigma_2^2$$
$$H_1: \sigma_1^2 \neq \sigma_2^2$$

To conduct the test, we select a random sample of observations, n_1, from one population and a random sample of observations, n_2, from the second population. The test statistic is defined as follows.

TEST STATISTIC FOR COMPARING TWO VARIANCES

$$F = \frac{s_1^2}{s_2^2} \qquad (12\text{–}1)$$

The terms s_1^2 and s_2^2 are the respective sample variances. If the null hypothesis is true, the test statistic follows the F distribution with $n_1 - 1$ and $n_2 - 1$ degrees of freedom. To reduce the size of the table of critical values, the *larger* sample variance is placed in the numerator; hence, the tabled F ratio is always larger than 1.00. Thus, the right-tail critical value is the only one required. The critical value of F for a two-tailed test is found by dividing the significance level in half ($\alpha/2$) and then referring to the appropriate degrees of freedom in Appendix B.6. An example will illustrate.

EXAMPLE

Lammers Limos offers limousine service from Government Center in downtown Toledo, Ohio, to Metro Airport in Detroit. Sean Lammers, president of the company, is considering two routes. One is via U.S. 25 and the other via I-75. He wants to study the time it takes to drive to the airport using each route and then compare the results. He collected the following sample data, which is reported in minutes. Using the .10 significance level, is there a difference in the variation in the driving times for the two routes?

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U.S. Route 25	Interstate 75
52	59
67	60
56	61
45	51
70	56
54	63
64	57
	65

SOLUTION

The mean driving times along the two routes are nearly the same. The mean time is 58.29 minutes for the U.S. 25 route and 59.0 minutes along the I-75 route. However, in evaluating travel times, Mr. Lammers is also concerned about the variation in the travel times. The first step is to compute the two sample variances. We'll use formula (3–9) to compute the sample standard deviations. To obtain the sample variances, we square the standard deviations.

U.S. ROUTE 25

$$\bar{x} = \frac{\Sigma x}{n} = \frac{408}{7} = 58.29 \qquad s = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}} = \sqrt{\frac{485.43}{7 - 1}} = 8.9947$$

INTERSTATE 75

$$\bar{x} = \frac{\Sigma x}{n} = \frac{472}{8} = 59.00 \qquad s = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}} = \sqrt{\frac{134}{8 - 1}} = 4.3753$$

There is more variation, as measured by the standard deviation, in the U.S. 25 route than in the I-75 route. This is consistent with his knowledge of the two routes; the U.S. 25 route contains more stoplights, whereas I-75 is a limited-access interstate highway. However, the I-75 route is several miles longer. It is important that the service

offered be both timely and consistent, so he decides to conduct a statistical test to determine whether there really is a difference in the variation of the two routes.

We use the six-step hypothesis test procedure.

Step 1: We begin by stating the null hypothesis and the alternate hypothesis. The test is two-tailed because we are looking for a difference in the variation of the two routes. We are *not* trying to show that one route has more variation than the other. For this example/solution, the subscript 1 indicates information for U.S. 25; the subscript 2 indicates information for I-75.

$$H_0: \sigma_1^2 = \sigma_2^2$$
$$H_1: \sigma_1^2 \neq \sigma_2^2$$

Step 2: We selected the .10 significance level.

Step 3: The appropriate test statistic follows the *F* distribution.

Step 4: The critical value is obtained from Appendix B.6, a portion of which is reproduced as Table 12–1. Because we are conducting a two-tailed test, the tabled significance level is .05, found by $\alpha/2 = .10/2 = .05$. There are $n_1 - 1 = 7 - 1 = 6$ degrees of freedom in the numerator and $n_2 - 1 = 8 - 1 = 7$ degrees of freedom in the denominator. To find the critical value, move horizontally across the top portion of the *F* table (Table 12–1 or Appendix B.6) for the .05 significance level to 6 degrees of freedom in the numerator. Then move down that column to the critical value opposite 7 degrees of freedom in the denominator. The critical value is 3.87. Thus, the decision rule is: Reject the null hypothesis if the ratio of the sample variances exceeds 3.87.

TABLE 12–1 Critical Values of the *F* Distribution, $\alpha = .05$

Degrees of Freedom for Denominator	Degrees of Freedom for Numerator			
	5	6	7	8
1	230	234	237	239
2	19.3	19.3	19.4	19.4
3	9.01	8.94	8.89	8.85
4	6.26	6.16	6.09	6.04
5	5.05	4.95	4.88	4.82
6	4.39	4.28	4.21	4.15
7	3.97	3.87	3.79	3.73
8	3.69	3.58	3.50	3.44
9	3.48	3.37	3.29	3.23
10	3.33	3.22	3.14	3.07

Step 5: Next we compute the ratio of the two sample variances, determine the value of the test statistic, and make a decision regarding the null hypothesis. Note that formula (12–1) refers to the sample *variances,* but we calculated the sample *standard deviations*. We need to square the standard deviations to determine the variances.

$$F = \frac{s_1^2}{s_2^2} = \frac{(8.9947)^2}{(4.3753)^2} = 4.23$$

The decision is to reject the null hypothesis because the computed *F* value (4.23) is larger than the critical value (3.87).

Step 6: We conclude there is a difference in the variation in the time to travel the two routes. Mr. Lammers will want to consider this in his scheduling.

The usual practice is to determine the F ratio by putting the larger of the two sample variances in the numerator. This will force the F ratio to be at least 1.00. This allows us to always use the right tail of the F distribution, thus avoiding the need for more extensive F tables.

A logical question arises: Is it possible to conduct one-tailed tests? For example, suppose in the previous example we suspected that the variance of the times using the U.S. 25 route, σ_1^2, is larger than the variance of the times along the I-75 route, σ_2^2. We would state the null and the alternate hypothesis as

$$H_0: \sigma_1^2 \leq \sigma_2^2$$
$$H_1: \sigma_1^2 > \sigma_2^2$$

The test statistic is computed as s_1^2/s_2^2. Notice that we labeled the population with the suspected large variance as population 1. So s_1^2 appears in the numerator. The F ratio will be larger than 1.00, so we can use the upper tail of the F distribution. Under these conditions, it is not necessary to divide the significance level in half. Because Appendix B.6 gives us only the .05 and .01 significance levels, we are restricted to these levels for one-tailed tests and .10 and .02 for two-tailed tests unless we consult a more complete table or use statistical software to compute the F statistic.

The Excel software has a procedure to perform a test of variances. Below is the output. The computed value of F is the same as that determined by using formula (12–1). The result of the one-tail hypothesis test is to reject the null hypothesis. The F of 4.23 is greater than the critical value of 3.87. Also, the p-value is less than 0.05. We conclude the variance of travel times on U.S. 25 is greater than the variance of travel times on I-75.

Variance Test

	A	B	C	D	E	F	G
1	U.S. 25	Interstate 75		F-Test Two-Sample for Variances			
2	52	59			*U.S. 25*	*Interstate 75*	
3	67	60		Mean	58.29	59.00	
4	56	61		Variance	80.90	19.14	
5	45	51		Observations	7.00	8.00	
6	70	56		df	6.00	7.00	
7	54	63		F	4.23		
8	64	57		P(F<=f) one-tail	0.04		
9		65		F Critical one-tail	3.87		
10							

SELF-REVIEW 12–1

Steele Electric Products Inc. assembles cell phones. For the last 10 days, Mark Nagy completed a mean of 39 phones per day, with a standard deviation of 2 per day. Debbie Richmond completed a mean of 38.5 phones per day, with a standard deviation of 1.5 per day. At the .05 significance level, can we conclude that there is more variation in Mark's daily production?

EXERCISES

1. What is the critical F value when the sample size for the numerator is six and the sample size for the denominator is four? Use a two-tailed test and the .10 significance level.
2. What is the critical F value when the sample size for the numerator is four and the sample size for the denominator is seven? Use a one-tailed test and the .01 significance level.
3. The following hypotheses are given.

$$H_0: \sigma_1^2 = \sigma_2^2$$
$$H_1: \sigma_1^2 \neq \sigma_2^2$$

A random sample of eight observations from the first population resulted in a standard deviation of 10. A random sample of six observations from the second population resulted in a standard deviation of 7. At the .02 significance level, is there a difference in the variation of the two populations?

4. The following hypotheses are given.

$$H_0: \sigma_1^2 \leq \sigma_2^2$$
$$H_1: \sigma_1^2 > \sigma_2^2$$

A random sample of five observations from the first population resulted in a standard deviation of 12. A random sample of seven observations from the second population showed a standard deviation of 7. At the .01 significance level, is there more variation in the first population?

5. Arbitron Media Research Inc. conducted a study of the iPod listening habits of men and women. One facet of the study involved the mean listening time. It was discovered that the mean listening time for a sample of 10 men was 35 minutes per day. The standard deviation was 10 minutes per day. The mean listening time for a sample of 12 women was also 35 minutes, but the standard deviation of the sample was 12 minutes. At the .10 significance level, can we conclude that there is a difference in the variation in the listening times for men and women?

6. A stockbroker at Critical Securities reported that the mean rate of return on a sample of 10 oil stocks was 12.6% with a standard deviation of 3.9%. The mean rate of return on a sample of 8 utility stocks was 10.9% with a standard deviation of 3.5%. At the .05 significance level, can we conclude that there is more variation in the oil stocks?

LO12-2
Use ANOVA to test a hypothesis that three or more population means are equal.

ANOVA: ANALYSIS OF VARIANCE

The *F* distribution is used to perform a wide variety of hypothesis tests. For example, when testing the equality of three or more population means, the Analysis of Variance (ANOVA) technique is used and the *F* statistic is used as the test statistic.

ANOVA Assumptions

The ANOVA to test the equality of three or more population means requires that three assumptions are true:

1. The populations follow the normal distribution.
2. The populations have equal standard deviations (σ).
3. The populations are independent.

When these conditions are met, *F* is used as the distribution of the test statistic.

Why do we need to study ANOVA? Why can't we just use the test of differences in population means discussed in the previous chapter? We could compare the population means two at a time. The major reason is the unsatisfactory buildup of Type I error. To explain further, suppose we have four different methods (A, B, C, and D) of training new recruits to be firefighters. We randomly assign each of the 40 recruits in this year's class to one of the four methods. At the end of the training program, we administer a test to measure understanding of firefighting techniques to the four groups. The question is: Is there a difference in the mean test scores among the four groups? An answer to this question will allow us to compare the four training methods.

Using the *t* distribution to compare the four population means, we would have to conduct six different *t* tests. That is, we would need to compare the mean scores for the four methods as follows: A versus B, A versus C, A versus D, B versus C, B versus D, and C versus D. For each *t* test, suppose we choose an $\alpha = .05$. Therefore, the probability of

a Type I error, rejecting the null when it is true, is .05. The complement is the probability of .95 that we do not reject the null when it is true. Because we conduct six separate (independent) tests, the probability that all six tests result in correct decisions is:

$$P(\textit{All correct}) = (.95)(.95)(.95)(.95)(.95)(.95) = .735$$

To find the probability of at least one error due to sampling, we subtract this result from 1. Thus, the probability of at least one incorrect decision due to sampling is 1 − .735 = .265. To summarize, if we conduct six independent tests using the *t* distribution, the likelihood of rejecting a true null hypothesis because of sampling error is an unsatisfactory .265. The ANOVA technique allows us to compare population means simultaneously at a selected significance level. It avoids the buildup of Type I error associated with testing many hypotheses.

ANOVA was first developed for applications in agriculture, and many of the terms related to that context remain. In particular, the term *treatment* is used to identify the different populations being examined. For example, treatment refers to how a plot of ground was treated with a particular type of fertilizer. The following illustration will clarify the term *treatment* and demonstrate an application of ANOVA.

EXAMPLE

Joyce Kuhlman manages a regional financial center. She wishes to compare the productivity, as measured by the number of customers served, among three employees. Four days are randomly selected and the number of customers served by each employee is recorded. The results are:

Wolfe	White	Korosa
55	66	47
54	76	51
59	67	46
56	71	48

SOLUTION

Is there a difference in the mean number of customers served? Chart 12–1 illustrates how the populations would appear if there were a difference in the treatment means. Note that the populations follow the normal distribution and the variation in each population is the same. However, the means are *not* the same.

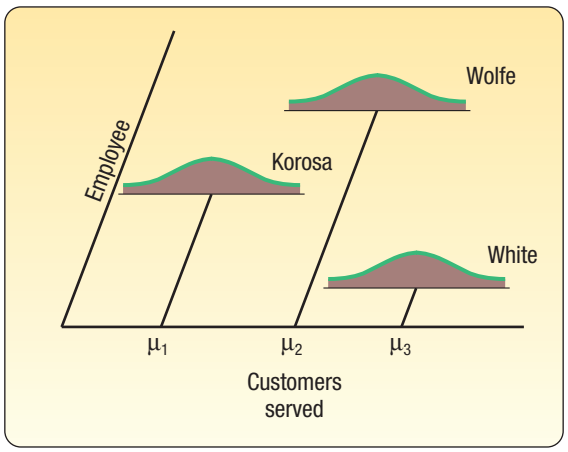


CHART 12–1 Case Where Treatment Means Are Different

Suppose there is no difference in the treatment means. This would indicate that the population means are the same. This is shown in Chart 12–2. Note again that the populations follow the normal distribution and the variation in each of the populations is the same.

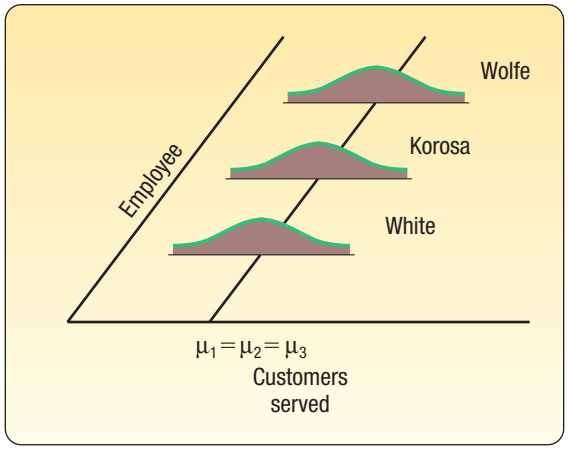


CHART 12–2 Case Where Treatment Means Are the Same

The ANOVA Test

How does the ANOVA test work? Recall that we want to determine whether the various sample means came from a single population or populations with different means. We actually compare these sample means through their variances. To explain, on page 392 we listed the assumptions required for ANOVA. One of those assumptions was that the standard deviations of the various normal populations had to be the same. We take advantage of this requirement in the ANOVA test. The underlying strategy is to estimate the population variance (standard deviation squared) two ways and then find the ratio of these two estimates. If this ratio is about 1, then logically the two estimates are the same, and we conclude that the population means are the same. If the ratio is quite different from 1, then we conclude that the population means are not the same. The *F* distribution serves as a referee by indicating when the ratio of the sample variances is too much greater than 1 to have occurred by chance.

Refer to the example/solution in the previous section. The manager wants to determine whether there is a difference in the mean number of customers served. To begin, find the overall mean of the 12 observations. It is 58, found by $(55 + 54 + \cdots + 48)/12$. Next, for each of the 12 observations find the difference between the particular value and the overall mean. Each of these differences is squared and these squares summed. This term is called the **total variation.**

TOTAL VARIATION The sum of the squared differences between each observation and the overall mean.

In our example, the total variation is 1,082, found by $(55 - 58)^2 + (54 - 58)^2 + \cdots + (48 - 58)^2$.

Next, break this total variation into two components: variation due to the **treatment variation** and **random variation.**

TREATMENT VARIATION The sum of the squared differences between each treatment mean and the grand or overall mean.

The variation due to treatments is also called variation between treatment means. In this example, we first square the difference between each treatment mean and the overall mean. The mean for Wolfe is 56 customers, found by (55 + 54 + 59 + 56)/4. The other means are 70 and 48, respectively. Then, each of the squared differences is multiplied by the number of observations in each treatment. In this case, the value is 4. Last, these values are summed together. This term is 992. The sum of the squares due to the treatments is:

$$4(56 - 58)^2 + 4(70 - 58)^2 + 4(48 - 58)^2 = 992$$

If there is considerable variation among the treatment means compared to the overall mean, it is logical that this term will be a large value. If the treatment means are similar, this value will be small. The smallest possible value would be zero. This would occur when all the treatment means are the same. In this case, all the treatment means would also equal the overall mean.

The other source of variation is referred to as **random variation,** or the error component.

RANDOM VARIATION The sum of the squared differences between each observation and its treatment mean.

In the example, this term is the sum of the squared differences between each value and the mean for each treatment or employee. This is also called the variation within the treatments. The error variation is 90.

$$(55 - 56)^2 + (54 - 56)^2 + \cdots + (48 - 48)^2 = 90$$

We determine the test statistic, which is the ratio of the two estimates of the population variance, from the following equation.

$$F = \frac{\text{Estimate of the population variance based on the differences between the treatment means}}{\text{Estimate of the population variance based on the variation within the treatments}}$$

Our first estimate of the population variance is based on the treatments, that is, the difference *between* the means. It is 992/2. Why did we divide by 2? Recall from Chapter 3, to find a sample variance [see formula (3–9)], we divide by the number of observations minus one. In this case, there are three treatments, so we divide by 2. Our first estimate of the population variance is 992/2.

The variance estimate *within* the treatments is the random variation divided by the total number of observations less the number of treatments—that is, 90/(12 – 3). Hence, our second estimate of the population variance is 90/9. This is actually a generalization of formula (11–4), we pooled the sample variances from two populations.

The last step is to take the ratio of these two estimates.

$$F = \frac{992/2}{90/9} = 49.6$$

Because this ratio is quite different from 1, we can conclude that the treatment means are not the same. There is a difference in the mean number of customers served by the three employees.

Here's another example, which deals with samples of different sizes.

EXAMPLE

Recently airlines cut services, such as meals and snacks during flights, and started charging for checked luggage. A group of four carriers hired Brunner Marketing Research Inc. to survey passengers regarding their level of satisfaction with a recent flight. The survey included questions on ticketing, boarding, in-flight service, baggage handling, pilot communication, and so forth. Twenty-five questions offered a range of possible answers: excellent, good, fair, or poor. A response of excellent was given a score of 4, good a 3, fair a 2, and poor a 1. These responses were then totaled, so the total score was an indication of the satisfaction with the flight. The greater the score, the higher the level of satisfaction with the service. The highest possible score was 100.

Brunner randomly selected and surveyed passengers from the four airlines. Below is the sample information. Is there a difference in the mean satisfaction level among the four airlines? Use the .01 significance level.

Northern	WTA	Pocono	Branson
94	75	70	68
90	68	73	70
85	77	76	72
80	83	78	65
	88	80	74
		68	65
		65	

SOLUTION

We will use the six-step hypothesis-testing procedure.

Step 1: State the null hypothesis and the alternate hypothesis. The null hypothesis is that the mean scores are the same for the four airlines.

$$H_0: \mu_N = \mu_W = \mu_P = \mu_B$$

The alternate hypothesis is that the mean scores are not all the same for the four airlines.

H_1: The mean scores are not all equal.

We can also think of the alternate hypothesis as "at least two mean scores are not equal."

If the null hypothesis is not rejected, we conclude that there is no difference in the mean scores for the four airlines. If H_0 is rejected, we conclude that there is a difference in at least one pair of mean scores, but at this point we do not know which pair or how many pairs differ.

Step 2: Select the level of significance. We selected the .01 significance level.

Step 3: Determine the test statistic. The test statistic follows the F distribution.

Step 4: Formulate the decision rule. To determine the decision rule, we need the critical value. The critical value for the F statistic is found in Appendix B.6. The critical values for the .05 significance level are found on the first page and the .01 significance level on the second page. To use this table, we need to know the degrees of freedom in the numerator and the denominator. The degrees of freedom in the numerator equal the number of treatments, designated as k, minus 1.

The degrees of freedom in the denominator are the total number of observations, *n,* minus the number of treatments. For this problem, there are four treatments and a total of 22 observations.

Degrees of freedom in the numerator $= k - 1 = 4 - 1 = 3$

Degrees of freedom in the denominator $= n - k = 22 - 4 = 18$

Refer to Appendix B.6 and the .01 significance level. Move horizontally across the top of the page to 3 degrees of freedom in the numerator. Then move down that column to the row with 18 degrees of freedom. The value at this intersection is 5.09. So the decision rule is to reject H_0 if the computed value of F exceeds 5.09.

Step 5: **Select the sample, perform the calculations, and make a decision.** It is convenient to summarize the calculations of the F statistic in an **ANOVA table.** The format for an ANOVA table is as follows. Statistical software packages also use this format.

ANOVA Table

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Treatments	SST	$k - 1$	$SST/(k - 1) = MST$	MST/MSE
Error	SSE	$n - k$	$SSE/(n - k) = MSE$	
Total	SS total	$n - 1$		

There are three values, or sum of squares, used to compute the test statistic F. You can determine these values by obtaining SS total and SSE, then finding SST by subtraction. The SS total term is the total variation, SST is the variation due to the treatments, and SSE is the variation within the treatments or the random error.

We usually start the process by finding SS total. This is the sum of the squared differences between each observation and the overall mean. The formula for finding SS total is:

$$\text{SS total} = \Sigma(x - \bar{x}_G)^2 \qquad \textbf{(12–2)}$$

where:

x is each sample observation.

$\bar{x}_G$ is the overall or grand mean.

Next determine SSE or the sum of the squared errors. This is the sum of the squared differences between each observation and its respective treatment mean. The formula for finding SSE is:

$$\text{SSE} = \Sigma(x - \bar{x}_c)^2 \qquad \textbf{(12–3)}$$

where:

$\bar{x}_c$ is the sample mean for treatment c.

The SSE is calculated:

$$\text{SSE} = \Sigma(x - \bar{x}_N)^2 + \Sigma(x - \bar{x}_W)^2 + \Sigma(x - \bar{x}_P)^2 + \Sigma(x - \bar{x}_B)^2$$

The detailed calculations of SS total and SSE for this example follow. To determine the values of SS total and SSE we start by calculating the overall or grand mean. There are 22 observations and the total is 1,664, so the grand mean is 75.64.

$$\bar{x}_G = \frac{1{,}664}{22} = 75.64$$

	Northern	WTA	Pocono	Branson	Total
	94	75	70	68	
	90	68	73	70	
	85	77	76	72	
	80	83	78	65	
		88	80	74	
			68	65	
			65		
Column total	349	391	510	414	1,664
n	4	5	7	6	22
Mean	87.25	78.20	72.86	69.00	75.64

Grand Mean

Next we find the deviation of each observation from the grand mean, square those deviations, and sum this result for all 22 observations. For example, the first sampled passenger had a score of 94 and the overall or grand mean is 75.64. So $(x - \bar{x}_G) = 94 - 75.64 = 18.36$. For the last passenger, $(x - \bar{x}_G) = 65 - 75.64 = -10.64$. The calculations for all other passengers follow.

Northern	WTA	Pocono	Branson
18.36	−0.64	−5.64	−7.64
14.36	−7.64	−2.64	−5.64
9.36	1.36	0.36	−3.64
4.36	7.36	2.36	−10.64
	12.36	4.36	−1.64
		−7.64	−10.64
		−10.64	

Then square each of these differences and sum all the values. Thus, for the first passenger:

$$(x - \bar{x}_G)^2 = (94 - 75.64)^2 = (18.36)^2 = 337.09$$

Finally, sum all the squared differences as formula (12–2) directs. Our SS total value is 1,485.10.

	Northern	WTA	Pocono	Branson	Total
	337.09	0.41	31.81	58.37	
	206.21	58.37	6.97	31.81	
	87.61	1.85	0.13	13.25	
	19.01	54.17	5.57	113.21	
		152.77	19.01	2.69	
			58.37	113.21	
			113.21		
Total	649.92	267.57	235.07	332.54	1,485.10

SS Total

To compute the term SSE, find the deviation between each observation and its treatment mean. In the example, the mean of the first treatment (that is, the passengers on Northern Airlines) is 87.25, found by $\bar{x}_N = 349/4$. The subscript *N* refers to Northern Airlines.

The first passenger rated Northern a 94, so $(x - \bar{x}_N) = (94 - 87.25) = 6.75$. The first passenger in the WTA group responded with a total score of 75, so $(x - \bar{x}_W) = (75 - 78.20) = -3.2$. The detail for all the passengers follows.

Northern	WTA	Pocono	Branson
6.75	−3.2	−2.86	−1
2.75	−10.2	0.14	1
−2.25	−1.2	3.14	3
−7.25	4.8	5.14	−4
	9.8	7.14	5
		−4.86	−4
		−7.86	

Each of these values is squared and then summed for all 22 observations. The four column totals can also be summed to find SSE. The values are shown in the following table.

	Northern	WTA	Pocono	Branson	Total
	45.5625	10.24	8.18	1	
	7.5625	104.04	0.02	1	
	5.0625	1.44	9.86	9	
	52.5625	23.04	26.42	16	
		96.04	50.98	25	SSE
			23.62	16	
			61.78		
Total	110.7500	234.80	180.86	68	594.41

So the SSE value is 594.41. That is, $\Sigma(x - \bar{x}_c)^2 = 594.41$.

Finally, we determine SST, the sum of the squares due to the treatments, by subtraction.

$$\text{SST} = \text{SS total} - \text{SSE} \qquad \textbf{(12–4)}$$

For this example:

$$\text{SST} = \text{SS total} - \text{SSE} = 1{,}485.10 - 594.41 = 890.69.$$

To find the computed value of *F*, work your way across the ANOVA table. The degrees of freedom for the numerator and the denominator are the same as in step 4 on page 396 when we were finding the critical value of *F*. The term **mean square** is another expression for an estimate of the variance. The mean square for treatments is SST divided by its degrees of freedom. The result is the **mean square for treatments** and is written MST. Compute the **mean square error** in a similar fashion. To be precise, divide SSE by its degrees of freedom. To complete the process and find *F*, divide MST by MSE.

Insert the particular values of *F* into an ANOVA table and compute the value of *F* as follows.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	*F*
Treatments	890.69	3	296.90	8.99
Error	594.41	18	33.02	
Total	1,485.10	21		

STATISTICS IN ACTION

Have you ever waited in line for a telephone and it seemed like the person using the phone talked on and on? There is evidence that people actually talk longer on public telephones when someone is waiting. In a recent survey, researchers measured the length of time that 56 shoppers in a mall spent on the phone (1) when they were alone, (2) when a person was using the adjacent phone, and (3) when a person was using an adjacent phone and someone was waiting to use the phone. The study, using the one-way ANOVA technique, showed that the mean time using the telephone was significantly less when the person was alone.

The computed value of F is 8.99, which is greater than the critical value of 5.09, so the null hypothesis is rejected.

Step 6: Interpret the result. We conclude the population means are not all equal. At this point, the results of the ANOVA only show that at least one pair of mean satisfaction scores are not the same among the four airlines. We cannot statistically show which airlines differ in satisfaction or which airlines have the highest or lowest satisfaction scores. The techniques for determining how the airlines differ are presented in the next section.

The calculations in the previous example/solution are tedious if the number of observations in each treatment is large. Many statistical software packages will perform the calculations and output the results. In the following illustration, Excel is used to calculate the descriptive statistics and ANOVA for the previous example/solution involving airlines and passenger ratings. There are some slight differences between the output and the previous calculations. These differences are due to rounding.

Airline Anova

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Northern	WTA	Pocono	Branson		Anova: Single Factor							
2	94	75	70	68									
3	90	68	73	70		SUMMARY							
4	85	77	76	72		*Groups*	*Count*	*Sum*	*Average*	*Variance*			
5	80	83	78	65		Northern	4	349	87.250	36.917			
6		88	80	74		WTA	5	391	78.200	58.700			
7			68	65		Pocono	7	510	72.857	30.143			
8			65			Branson	6	414	69.000	13.600			
9													
10						ANOVA							
11						*Source of Variation*	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*	
12						Between Groups	890.684	3	296.895	8.99	0.0007	3.160	
13						Within Groups	594.407	18	33.023				
14													
15						Total	1485.091	21					
16													

Notice Excel uses the term "Between Groups" for treatments and "Within Groups" for error. However, they have the same meanings. The p-value is .0007. This is the probability of finding a value of the test statistic this large or larger when the null hypothesis is true. To put it another way, it is the likelihood of calculating an F value larger than 8.99 with 3 degrees of freedom in the numerator and 18 degrees of freedom in the denominator. So when we reject the null hypothesis in this instance, there is a very small likelihood of committing a Type I error!

SELF-REVIEW 12–2

Citrus Clean is a new all-purpose cleaner being test-marketed by placing displays in three different locations within various supermarkets. The number of 12-ounce bottles sold from each location within the supermarket is reported below.

Near Bread	Near Beer	With Cleaners
18	12	26
14	18	28
19	10	30
17	16	32

At the .05 significance level, is there a difference in the mean number of bottles sold at the three locations?

(a) State the null hypothesis and the alternate hypothesis.
(b) What is the decision rule?
(c) Compute the values of SS total, SST, and SSE.
(d) Develop an ANOVA table.
(e) What is your decision regarding the null hypothesis?

EXERCISES

7. The following are four observations collected from each of three treatments. Test the hypothesis that the treatment means are equal. Use the .05 significance level.

Treatment 1	Treatment 2	Treatment 3
8	3	3
6	2	4
10	4	5
9	3	4

   a. State the null and the alternate hypotheses.
   b. What is the decision rule?
   c. Compute SST, SSE, and SS total.
   d. Complete an ANOVA table.
   e. State your decision regarding the null hypothesis.

8. The following are six observations collected from treatment 1, four observations collected from treatment 2, and five observations collected from treatment 3. Test the hypothesis at the .05 significance level that the treatment means are equal.

Treatment 1	Treatment 2	Treatment 3
9	13	10
7	20	9
11	14	15
9	13	14
12		15
10		

   a. State the null and the alternate hypotheses.
   b. What is the decision rule?
   c. Compute SST, SSE, and SS total.
   d. Complete an ANOVA table.
   e. State your decision regarding the null hypothesis.

9. FILE A real estate developer is considering investing in a shopping mall on the outskirts of Atlanta, Georgia. Three parcels of land are being evaluated. Of particular importance is the income in the area surrounding the proposed mall. A random sample of four families is selected near each proposed mall. Following are the sample results. At the .05 significance level, can the developer conclude there is a difference in the mean income? Use the usual six-step hypothesis testing procedure.

Southwyck Area ($000)	Franklin Park ($000)	Old Orchard ($000)
64	74	75
68	71	80
70	69	76
60	70	78

10. FILE The manager of a computer software company wishes to study the number of hours per week senior executives by type of industry spend at their desktop computers. The manager selected a sample of five executives from each of three

industries. At the .05 significance level, can she conclude there is a difference in the mean number of hours spent per week by industry?

Banking	Retail	Insurance
32	28	30
30	28	28
30	26	26
32	28	28
30	30	30

LO12-3
Use confidence intervals to test and interpret differences between pairs of population means.

INFERENCES ABOUT PAIRS OF TREATMENT MEANS

Suppose we carry out the ANOVA procedure, make the decision to reject the null hypothesis, and conclude that all the treatment means are not the same. Sometimes we may be satisfied with this conclusion, but in other instances we may want to know which treatment means differ. This section provides the details for this analysis.

Recall in the previous example/solution regarding airline passenger ratings, we concluded that there was a difference in the treatment means. That is, the null hypothesis was rejected and the alternate hypothesis accepted. The conclusion is that at least one of the airline's mean level of satisfaction is different for the others. Now, the question is which of the four airlines differ?

Several procedures are available to answer this question. The simplest is through the use of confidence intervals, that is, formula (9–2). From the computer output of the example on page 400, the sample mean score for those passengers rating Northern's service is 87.25, and for those rating Branson's service, the sample mean score is 69.00. Is there enough disparity to justify the conclusion that there is a significant difference in the mean satisfaction scores of the two airlines?

The t distribution, described in Chapters 10 and 11, is used as the basis for this test. Recall that one of the assumptions of ANOVA is that the population variances are the same for all treatments. This common population value is the **mean square error,** or MSE, and is determined by SSE/($n - k$). A confidence interval for the difference between two populations is found by:

CONFIDENCE INTERVAL FOR THE DIFFERENCE IN TREATMENT MEANS

$$(\bar{x}_1 - \bar{x}_2) \pm t\sqrt{\text{MSE}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \qquad \textbf{(12–5)}$$

where:

$\bar{x}_1$ is the mean of the first sample.
$\bar{x}_2$ is the mean of the second sample.
t is obtained from Appendix B.5. The degrees of freedom are equal to $n - k$.
MSE is the mean square error term obtained from the ANOVA table [SSE/($n - k$)].
n_1 is the number of observations in the first sample.
n_2 is the number of observations in the second sample.

How do we decide whether there is a difference in the treatment means? If the confidence interval includes zero, there is *not* a difference between the treatment means. For example, if the left endpoint of the confidence interval has a negative sign and the right endpoint has a positive sign, the interval includes zero and the two means do not differ. So if we develop a confidence interval from formula (12–5) and find the difference in the sample means was 5.00—that is, if $\bar{x}_1 - \bar{x}_2 = 5$ and

$t\sqrt{\text{MSE}\left(\frac{1}{n_1}+\frac{1}{n_2}\right)} = 12$—the confidence interval would range from −7.00 up to 17.00. To put it in symbols:

$$(\bar{x}_1 - \bar{x}_2) \pm t\sqrt{\text{MSE}\left(\frac{1}{n_1}+\frac{1}{n_2}\right)} = 5.00 \pm 12.00 = -7.00 \text{ up to } 17.00$$

Note that zero is in this interval. Therefore, we conclude that there is no significant difference in the selected treatment means.

On the other hand, if the endpoints of the confidence interval have the same sign, this indicates that the treatment means differ. For example, if $\bar{x}_1 - \bar{x}_2 = -0.35$ and $t\sqrt{\text{MSE}\left(\frac{1}{n_1}+\frac{1}{n_2}\right)} = 0.25$, the confidence interval would range from −0.60 up to −0.10. Because −0.60 and −0.10 have the same sign, both negative, zero is not in the interval and we conclude that these treatment means differ.

Using the previous airline example, let us compute the confidence interval for the difference between the mean scores of passengers on Northern and Branson. With a 95% level of confidence, the endpoints of the confidence interval are 10.457 and 26.043.

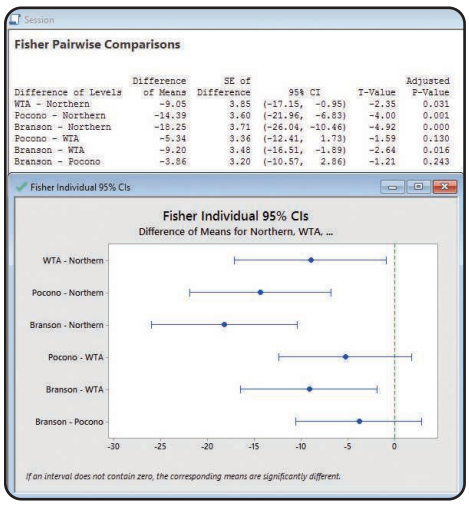
$$(\bar{x}_N - \bar{x}_B) \pm t\sqrt{\text{MSE}\left(\frac{1}{n_N}+\frac{1}{n_B}\right)} = (87.25 - 69.00) \pm 2.101\sqrt{33.023\left(\frac{1}{4}+\frac{1}{6}\right)}$$
$$= 18.25 \pm 7.793$$

where:

$\bar{x}_N$ is 87.25.
$\bar{x}_B$ is 69.00.
t is 2.101: from Appendix B.5 with $(n - k) = 22 - 4 = 18$ degrees of freedom.
MSE is 33.023: from the ANOVA table with SSE/$(n - k)$ = 594.4/18.
n_N is 4.
n_B is 6.

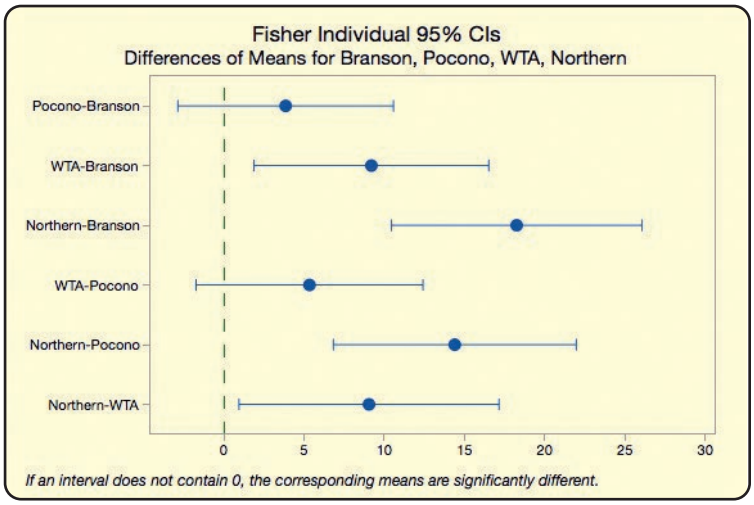
The 95% confidence interval ranges from 10.457 up to 26.043. Both endpoints are positive; hence, we can conclude these treatment means differ significantly. That is, passengers on Northern Airlines rated service significantly different from those on Branson Airlines.

The confidence intervals for the differences between each pair of means can be obtained directly using statistical software. The following confidence intervals were computed using the one-way ANOVA in Minitab. Statistical software, such as Minitab, offers a variety of methods to control Type I error when making multiple comparisons. The following analysis used Fisher's method to compare means.



Fisher Pairwise Comparisons

Difference of Levels	Difference of Means	SE of Difference	95% CI	T-Value	Adjusted P-Value
WTA - Northern	-9.05	3.85	(-17.15, -0.95)	-2.35	0.031
Pocono - Northern	-14.39	3.60	(-21.96, -6.83)	-4.00	0.001
Branson - Northern	-18.25	3.71	(-26.04, -10.46)	-4.92	0.000
Pocono - WTA	-5.34	3.36	(-12.41, 1.73)	-1.59	0.130
Branson - WTA	-9.20	3.48	(-16.51, -1.89)	-2.64	0.016
Branson - Pocono	-3.86	3.20	(-10.57, 2.86)	-1.21	0.243



The output shows the confidence intervals for the difference between each pair of treatment means. The first row shows the confidence interval that compares WTA and Northern. It shows a confidence interval that does not include zero. It also shows the *p*-value for a hypothesis test that the means of WTA and Northern are equal. The hypothesis is rejected because a *p*-value of 0.031 is less than an assumed α of 0.05. Both results indicate that the WTA and Northern means are significantly different. Reviewing the entire table, only two pairs of means are not significantly different: Pocono and WTA, and Branson and Pocono. All other confidence intervals do not include zero and have *p*-values less than 0.05. Therefore, all other pairs of means are significantly different.

The graphic illustrates the results of the confidence interval analysis. Each confidence interval is represented by its endpoints and treatment mean. Note that a difference of zero is illustrated with the vertical dotted line. Two of the intervals include zero, Pocono and WTA, and Branson and Pocono. The others do not include zero so the means are significantly different. The following pairs of means are different: WTA and Northern, Pocono and Northern, Branson and Northern, and Branson and WTA.

We should emphasize that this investigation is a step-by-step process. The initial step is to conduct the ANOVA test. Only if the null hypothesis that the treatment means are equal is rejected should any analysis of the individual treatment means be attempted.

SELF-REVIEW 12–3

The following data are the semester tuition charges ($000) for a sample of five private colleges in the Northeast region of the United States, four in the Southeast region, and five in the West region. At the .05 significance level, can we conclude there is a difference in the mean tuition rates for the various regions?

Northeast ($000)	Southeast ($000)	West ($000)
40	38	37
41	39	38
42	40	36
40	38	37
42		36

(a) State the null and the alternate hypotheses.
(b) What is the decision rule?
(c) Develop an ANOVA table. What is the value of the test statistic?
(d) What is your decision regarding the null hypothesis?
(e) Could there be a significant difference between the mean tuition in the Northeast and that of the West? If so, develop a 95% confidence interval for that difference.

EXERCISES

11. FILE The following are three observations collected from treatment 1, five observations collected from treatment 2, and four observations collected from treatment 3. Test the hypothesis that the treatment means are equal at the .05 significance level.

Treatment 1	Treatment 2	Treatment 3
8	3	3
11	2	4
10	1	5
	3	4
	2	

a. State the null hypothesis and the alternate hypothesis.
b. What is the decision rule?
c. Compute SST, SSE, and SS total.
d. Complete an ANOVA table.
e. State your decision regarding the null hypothesis.
f. If H_0 is rejected, can we conclude that treatment 1 and treatment 2 differ? Use the 95% level of confidence.

12. FILE The following are six observations collected from treatment 1, ten observations collected from treatment 2, and eight observations collected from treatment 3. Test the hypothesis that the treatment means are equal at the .05 significance level.

Treatment 1	Treatment 2	Treatment 3
3	9	6
2	6	3
5	5	5
1	6	5
3	8	5
1	5	4
	4	1
	7	5
	6	
	4	

a. State the null hypothesis and the alternate hypothesis.
b. What is the decision rule?
c. Compute SST, SSE, and SS total.
d. Complete an ANOVA table.
e. State your decision regarding the null hypothesis.
f. If H_0 is rejected, can we conclude that treatment 2 and treatment 3 differ? Use the 95% level of confidence.

13. A senior accounting major at Midsouth State University has job offers from four CPA firms. To explore the offers further, she asked a sample of recent trainees how many months each worked for the firm before receiving a raise in salary. The sample information is submitted to Minitab with the following results:

```
Analysis of Variance
Source     DF       SS       MS       F       P
Factor      3    32.33    10.78    2.36   0.133
Error      10    45.67     4.57
Total      13    78.00
```

At the .05 level of significance, is there a difference in the mean number of months before a raise was granted among the four CPA firms?

14. FILE A stock analyst wants to determine whether there is a difference in the mean return on equity for three types of stock: utility, retail, and banking stocks. The following output is obtained:

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Factor	2	303.697	151.848	5.50	0.0202
Error	12	331.381	27.615		
Total	14	635.077			

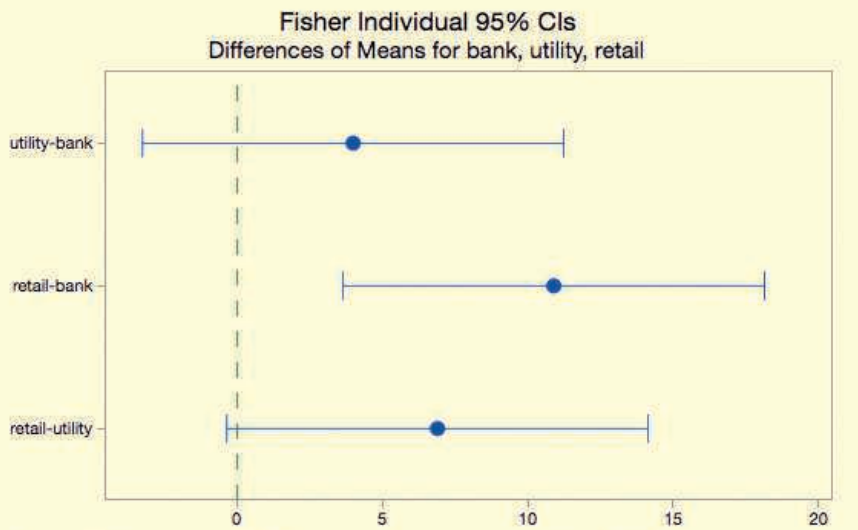
Means

Factor	N	Mean	StDev	95% CI
bank	5	5.318	3.904	(0.198, 10.438)
utility	5	9.3160	1.5376	(4.1956, 14.4364)
retail	5	16.212	8.077	(11.092, 21.332)

Pooled StDev = 5.25500

Fisher Individual Tests for Differences of Means

Difference of Levels	Difference of Means	SE of Difference	95% CI	T-Value	Adjusted P-Value
utility-bank	3.998	3.324	(-3.243, 11.239)	1.20	0.2522
retail-bank	10.894	3.324	(3.653, 18.135)	3.28	0.0066
retail-utility	6.896	3.324	(-0.345, 14.137)	2.07	0.0602



If an interval does not contain 0, the corresponding means are significantly different.

a. Using the .05 level of significance, is there a difference in the mean return on equity among the three types of stock?

b. Can the analyst conclude there is a difference between the mean return on equity for utility and retail stocks? For utility and banking stocks? For banking and retail stocks? Explain.

LO12-4

Use a blocking variable in a two-way ANOVA to test a hypothesis that three or more population means are equal.

TWO-WAY ANALYSIS OF VARIANCE

In the example/solution in the previous section, we divided the total variation in passenger ratings of the airlines into two categories: the variation between the treatments and the variation within the treatments. We also called the variation within the treatments the error or the random variation. To put it another way, we considered only two sources of variation: that due to the treatments and the random differences. In the airline passenger ratings example, there may be other causes of variation. These factors might

include, for example, the season of the year, the particular airport, or the number of passengers on the flight.

The benefit of considering other factors is that we can reduce the error variance. That is, if we can reduce the denominator of the F statistic (reducing the error variance or, more directly, the SSE term), the value of F will be larger, causing us to reject the hypothesis of equal treatment means. In other words, if we can explain more of the variation, then there is less "error." An example will clarify the reduction in the error variance.

EXAMPLE

WARTA, the Warren Area Regional Transit Authority, is expanding bus service from the suburb of Starbrick into the central business district of Warren. There are four routes being considered from Starbrick to downtown Warren: (1) via U.S. 6, (2) via the West End, (3) via the Hickory Street Bridge, and (4) via Route 59. WARTA conducted several tests to determine whether there was a difference in the mean travel times along the four routes. Because there will be many different drivers, the test was set up so each driver drove along each of the four routes. Below is the travel time, in minutes, for each driver–route combination.

© John A. Rizzo/Getty Images RF

	Travel Time from Starbrick to Warren (minutes)			
Driver	**U.S. 6**	**West End**	**Hickory St.**	**Rte. 59**
Deans	18	17	21	22
Snaverly	16	23	23	22
Ormson	21	21	26	22
Zollaco	23	22	29	25
Filbeck	25	24	28	28

At the .05 significance level, is there a difference in the mean travel time along the four routes? If we remove the effect of the drivers, is there a difference in the mean travel time?

SOLUTION

To begin, we conduct a test of hypothesis using a one-way ANOVA. That is, we consider only the four routes. Under this condition, differences in travel times are due to either treatment or random variation. In this example/solution, the subscripts correspond to the treatments or routes: 1 for U.S. 6, 2 for West End, 3 for Hickory Street, and 4 for Route 59. The null hypothesis and the alternate hypothesis for comparing the mean travel time along the four routes are:

H_0: $\mu_1 = \mu_2 = \mu_3 = \mu_4$
H_1: Not all treatment means are the same.

There are four routes, so the numerator degrees of freedom is $(k - 1) = (4 - 1) = 3$. There are 20 observations, so the degrees of freedom in the denominator is

$(n - k) = (20 - 4) = 16$. From Appendix B.6, at the .05 significance level, the critical value of F is 3.24. The decision rule is to reject the null hypothesis if the computed F test statistic's value is greater than 3.24.

We use Excel to perform the calculations and output the results. The computed value of F is 2.483, so we decide to not reject the null hypothesis. We conclude there is no difference in the mean travel time along the four routes. There is no reason to conclude that any one of the routes is faster than any other.

Drivers and Routes.xlsx

	A	B	C	D	E
1					
2		Routes			
3	Driver	U.S 6	West End	Hickory St.	Route 59
4	Deans	18	17	21	22
5	Snaverly	16	23	23	22
6	Ormson	21	21	26	22
7	Zollaco	23	22	29	25
8	Filbeck	25	24	28	28

Anova: Single Factor

SUMMARY

Groups	Count	Sum	Average	Variance
U.S 6	5	103	20.6	13.3
West End	5	107	21.4	7.3
Hickory St.	5	127	25.4	11.3
Route 59	5	119	23.8	7.2

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	72.8	3	24.267	2.483	0.098	3.239
Within Groups	156.4	16	9.775			
Total	229.2	19				

Treatment (Block)

Error

From the above Excel output, the mean travel times along the routes were 20.6 minutes along U.S. 6, 21.4 minutes along the West End route, 25.4 minutes using Hickory Street, and 23.8 minutes using Route 59. We conclude these differences could reasonably be attributed to chance. From the ANOVA table, we note SST is 72.8, SSE is 156.4, and SS total is 229.2.

In this example, we only considered the variation due to the treatments (routes) and took all the remaining variation to be random. If we include the effect or variance of the drivers, this would allow us to reduce the SSE term, and the computed values of the F statistics would be larger.

In this case, we let the drivers be the **blocking variable.** To include the variance due to the drivers, we need to determine the sum of squares due to the blocks. In a two-way ANOVA, the sum of squares due to blocks is found by the following formula.

BLOCKING VARIABLE A second treatment variable that when included in the ANOVA analysis will have the effect of reducing the SSE term.

$$\text{SSB} = k\Sigma(\bar{x}_b - \bar{x}_G)^2 \qquad \textbf{(12–6)}$$

where:

k is the number of treatments.
b is the number of blocks.
$\bar{x}_b$ is the sample mean of block b.
$\bar{x}_G$ is the overall or grand mean.

From the calculations on the next page, the means for the respective drivers are 19.5 minutes, 21 minutes, 22.5 minutes, 24.75 minutes, and 26.25 minutes. The overall mean is 22.8 minutes, found by adding the travel time for all 20 drives (456 minutes) and dividing by 20.

Travel Time from Starbrick to Warren (minutes)						
Driver	**U.S. 6**	**West End**	**Hickory St.**	**Rte. 59**	**Driver Sums**	**Driver Means**
Deans	18	17	21	22	78	19.50
Snaverly	16	23	23	22	84	21.00
Ormson	21	21	26	22	90	22.50
Zollaco	23	22	29	25	99	24.75
Filbeck	25	24	28	28	105	26.25

Substituting this information into formula (12–6) we determine SSB, the sum of squares due to the drivers (the blocking variable), is 119.7.

$$\begin{aligned}\text{SSB} &= k\Sigma(\bar{x}_b - \bar{x}_G)^2 \\ &= 4(19.5 - 22.8)^2 + 4(21.0 - 22.8)^2 + 4(22.5 - 22.8)^2 \\ &\quad + 4(24.75 - 22.8)^2 + 4(26.25 - 22.8)^2 \\ &= 119.7\end{aligned}$$

The SSE term is found by subtraction.

SUM OF SQUARES ERROR, TWO-WAY $\text{SSE} = \text{SS total} - \text{SST} - \text{SSB}$ **(12–7)**

The same format is used in the two-way ANOVA table as in the one-way case, except there is an additional row for the blocking variable. SS total and SST are calculated as before, and SSB is found from formula (12–6). The values for the various components of the ANOVA table are computed as follows.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	*F*
Treatments	SST	$k - 1$	$\text{SST}/(k - 1) = \text{MST}$	MST/MSE
Blocks	SSB	$b - 1$	$\text{SSB}/(b - 1) = \text{MSB}$	MSB/MSE
Error	SSE	$(k - 1)(b - 1)$	$\text{SSE}/[(k - 1)(b - 1)] = \text{MSE}$	
Total	SS total	$n - 1$		

SSE is found by formula (12–7).

$$\text{SSE} = \text{SS total} - \text{SST} - \text{SSB} = 229.2 - 72.8 - 119.7 = 36.7$$

Source of Variation	(1) Sum of Squares	(2) Degrees of Freedom	(3) Mean Square (1)/(2)
Treatments	72.8	3	24.27
Blocks	119.7	4	29.93
Error	36.7	12	3.06
Total	229.2	19	

There is disagreement at this point. If the purpose of the blocking variable (the drivers in this example) was only to reduce the error variation, we should not conduct a test of hypothesis for the difference in block means. That is, if our goal was to reduce the MSE term, then we should not test a hypothesis regarding the blocking variable. On the other hand, we may wish to give the blocks the same status as the treatments and conduct a hypothesis test. In the latter case, when the blocks are important enough to be considered as a second factor, we refer to this as a **two-factor experiment.** In many cases the decision is not clear. In our example we are concerned about the difference in

the travel time for the different drivers, so we will conduct the hypothesis test of equal block means. The subscripts are the first letter of each driver's name. The two sets of hypotheses are:

1. H_0: The treatment means are equal ($\mu_1 = \mu_2 = \mu_3 = \mu_4$).
   H_1: At least one treatment mean is different.
2. H_0: The block means are equal ($\mu_D = \mu_S = \mu_O = \mu_Z = \mu_F$).
   H_1: At least one block mean is different.

First, we will test the hypothesis concerning the treatment means. There are $(k - 1) = (4 - 1) = 3$ degrees of freedom in the numerator and $(b - 1)(k - 1) = (5 - 1)(4 - 1) = 12$ degrees of freedom in the denominator. Using the .05 significance level, the critical value of F is 3.49. The null hypothesis that the mean times for the four routes are the same is rejected if the F ratio exceeds 3.49.

$$F = \frac{\text{MST}}{\text{MSE}} = \frac{24.27}{3.06} = 7.93$$

The null hypothesis is rejected and we conclude that at least one of the route's mean travel times is different from the other routes. WARTA will want to conduct some tests to determine which treatment means differ.

Next, we test to find whether the travel times for the various drivers are equal. The degrees of freedom in the numerator for blocks are $(b - 1) = (5 - 1) = 4$. The degrees of freedom for the denominator are the same as before: $(b - 1)(k - 1) = (5 - 1)(4 - 1) = 12$. The null hypothesis that the block means are the same is rejected if the F ratio exceeds 3.26.

$$F = \frac{\text{MSB}}{\text{MSE}} = \frac{29.93}{3.06} = 9.78$$

The null hypothesis about the block means is rejected, and we conclude that at least one driver's mean travel time is different from the other drivers. Thus, WARTA management can conclude, based on the sample results, that there is a difference in the mean travel times of drivers.

The Excel spreadsheet has a two-factor ANOVA procedure. The output for the WARTA example just completed follows. This output also includes the p-values. The p-value for the null hypothesis regarding the drivers is .001 and .004 for the routes. These p-values confirm that the null hypotheses for treatments and blocks should both be rejected because the p-value is less than the significance level.

Drivers and Routes.xlsx

	Routes			
Driver	U.S 6	West End	Hickory St.	Route 59
Deans	18	17	21	22
Snaverly	16	23	23	22
Ormson	21	21	26	22
Zollaco	23	22	29	25
Filbeck	25	24	28	28

Anova: Two-Factor Without Replication

SUMMARY	Count	Sum	Average	Variance
Deans	4	78	19.50	5.67
Snaverly	4	84	21.00	11.33
Ormson	4	90	22.50	5.67
Zollaco	4	99	24.75	9.58
Filbeck	4	105	26.25	4.25
U.S 6	5	103	20.60	13.30
West End	5	107	21.40	7.30
Hickory St.	5	127	25.40	11.30
Route 59	5	119	23.80	7.20

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Rows	119.7	4	29.925	9.785	0.001	3.259
Columns	72.8	3	24.267	7.935	0.004	3.490
Error	36.7	12	3.058			
Total	229.2	19				

Block (Driver)

Treatment (Route)

SELF-REVIEW 12–4

Vive Shampoo sells three shampoos, one each for dry, normal, and oily hair. Sales, in millions of dollars, for the past 5 months are given in the following table. Using the .05 significance level, test whether the mean sales differ for the three types of shampoo or by month.

	Sales ($ million)		
Month	**Dry**	**Normal**	**Oily**
June	7	9	12
July	11	12	14
August	13	11	8
September	8	9	7
October	9	10	13

EXERCISES

For exercises 15 and 16, conduct a test of hypothesis to determine whether the block or the treatment means differ. Using the .05 significance level: (a) state the null and alternate hypotheses for treatments; (b) state the decision rule for treatments; and (c) state the null and alternate hypotheses for blocks. Also, state the decision rule for blocks, then: (d) compute SST, SSB, SS total, and SSE; (e) complete an ANOVA table; and (f) give your decision regarding the two sets of hypotheses and interpret the results.

15. The following data was collected for a two-factor ANOVA with two treatments and three blocks.

	Treatment	
Block	**1**	**2**
A	46	31
B	37	26
C	44	35

16. The following data was collected for a two-factor ANOVA with three treatments and three blocks.

	Treatment		
Block	**1**	**2**	**3**
A	12	14	8
B	9	11	9
C	7	8	8

17. FILE Chapin Manufacturing Company operates 24 hours a day, 5 days a week. The workers rotate shifts each week. Management is interested in whether there is a difference in the number of units produced when the employees work on various shifts. A sample of five workers is selected and their output recorded on each shift. At the .05 significance level, can we conclude there is a difference in the mean production rate by shift or by employee?

	Units Produced		
Employee	**Day**	**Afternoon**	**Night**
Skaff	31	25	35
Lum	33	26	33
Clark	28	24	30
Treece	30	29	28
Morgan	28	26	27

18. FILE There are three hospitals in the Tulsa, Oklahoma, area. The following data show the number of outpatient surgeries performed on Monday, Tuesday, Wednesday, Thursday, and Friday at each hospital last week. At the .05 significance level, can we conclude there is a difference in the mean number of surgeries performed by hospital or by day of the week?

	Number of Surgeries Performed		
Day	**St. Luke's**	**St. Vincent**	**Mercy**
Monday	14	18	24
Tuesday	20	24	14
Wednesday	16	22	14
Thursday	18	20	22
Friday	20	28	24

LO12-5
Perform a two-way ANOVA with interaction and describe the results.

TWO-WAY ANOVA WITH INTERACTION

In the previous section, we studied the separate or independent effects of two variables, or factors, on a response variable, travel time. In the example/solution, the two factors were the bus routes and the drivers and the response was travel time. The analysis shows two significant results. First, the mean travel times between routes, averaged over all drivers, are different. Second, the mean travel times between the five drivers, averaged over all the routes, are different. What could explain these differences? The differences between routes may simply be related to differences in the distance of the routes. We really didn't study the distance of the various routes. Perhaps the differences are explained by how fast, on average, the drivers drive regardless of the route.

There is another effect that may influence travel time that we have not considered. This is called the **interaction effect** between route and driver on travel time. That is, differences in travel time may depend on both the driver and the route. For example, it is possible that one of the drivers is especially good driving one of the routes. Perhaps one driver knows how to effectively time the traffic lights or how to avoid heavily congested intersections for one or more of the routes. In this case, differences in mean travel time may depend on the combined effect of driver and route. The results of the interaction of driver and route can provide interesting information.

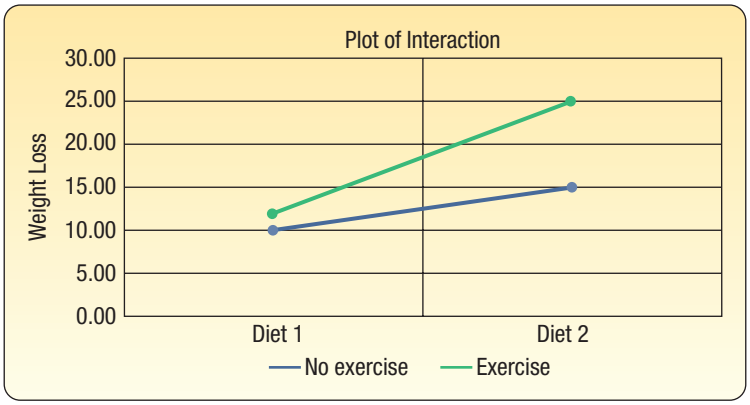
INTERACTION The effect of one factor on a response variable differs depending on the value of another factor.

Interaction Plots

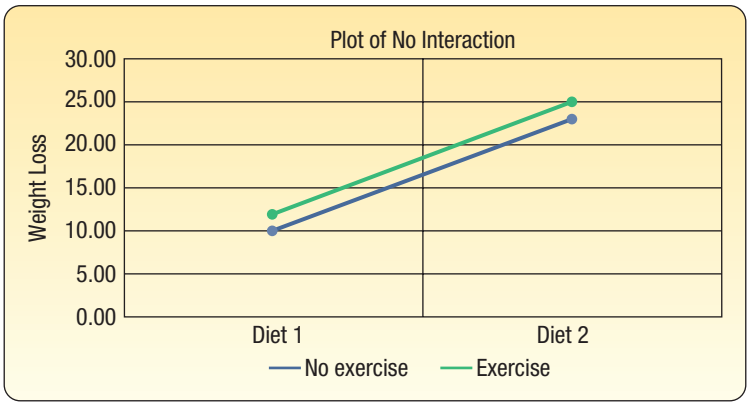
An everyday illustration of interaction is the effect of diet and exercise on body weight. It is generally agreed that body weight (the response variable) can be affected by two factors, diet and exercise. However, research shows there is also a combined or *interaction* effect of diet and exercise on weight loss. That is, the amount of weight loss will be different and depend on diet AND whether people exercise.

The following graph, or interaction plot, illustrates the interaction of diet and exercise. First, for people who do not exercise, the mean weight losses for Diet 1 and Diet 2 are plotted. These are connected with the blue line. Clearly, there is a difference in weight loss for the two diets. Second, for people who did exercise, the mean weight losses for Diet 1 and Diet 2 are plotted. These are connected with the green line. Again, there is a clear difference in weight loss between Diet 1 and Diet 2 for people who exercise. The plot also shows an interaction effect between diet and exercise on weight

loss. Notice the two lines are not parallel to each other. For Diet 1, the mean weight loss is more when people also exercise. For Diet 2, the mean weight loss is also more when people also exercise, but the weight loss is much greater than for Diet 1. So, what is the effect of diet and exercise on weight loss? It depends on the combined, or interaction, effects of diet and exercise.



How would the interaction plot look if there were no interaction? The following graph shows an analysis of diet and exercise with no interaction.



In this case the lines are parallel. Comparing the means, the effect of exercise on weight loss for Diet 1 and Diet 2 is the same. The estimated weight loss is about two pounds. In addition, the effect of diet is the same whether people exercise or not. It is about 13 pounds.

Testing for Interaction

To test for an interaction effect, we use a two-way ANOVA with interaction. To illustrate, we return to the previous WARTA example/solution. Restating the issue facing WARTA management: They want to expand bus service from downtown Warren to Starbrick. So far they have concluded, based on statistical analysis, that there is a difference in the mean travel time along the four proposed routes and a difference in the mean travel times of the five drivers. But it is possible that the combination, or the interaction between routes and drivers, has a significant effect on mean travel time.

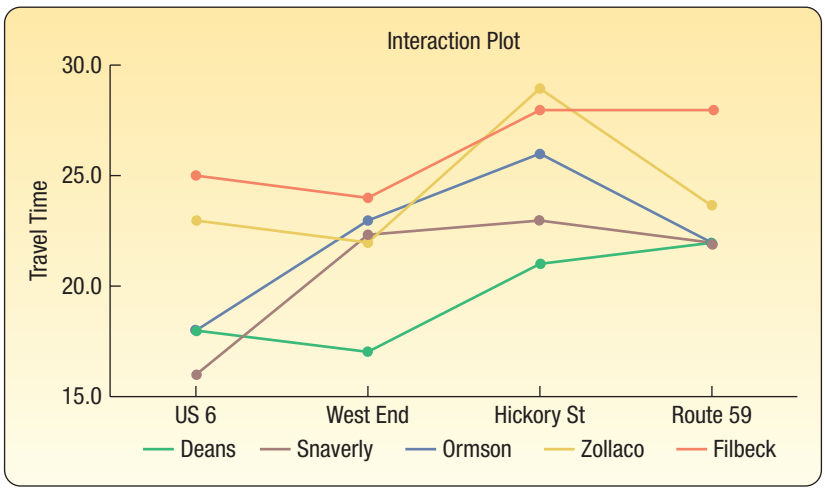
In this analysis, we call the two variables, route and driver, **factors.** We refer to the variable, travel time, as the **response** variable. To test for interaction, the sample data must be replicated for each route. In this case, each driver drives each route three times so there are three observed times for each route/driver combination. This information is summarized in the following Excel spreadsheet.

		Routes			Means for Drivers	
	US 6	West End	Hickory St	Route 59		
Deans	18	17	21	22		
	15	14	20	19		
	21	20	22	25	19.50	
Snaverly	16	23	23	22		
	19	19	24	20		
	13	25	22	24	20.83	
Ormson	21	21	26	22		
	19	23	24	24		
	14	25	28	20	22.25	
Zollaco	23	22	29	25		
	21	24	30	20		
	25	20	28	26	24.42	
Filbeck	25	24	28	28		
	24	25	29	30		Grand Mean
	26	23	27	26	26.25	
Means for Routes	20.00	21.67	25.40	23.53	22.65	

To evaluate interaction effects, a useful first step is to plot the means for each driver/route combination. For the driver/route combination Driver Deans using Route 6, the mean is 18 minutes, found by (18 + 15 + 21)/3. For the driver/route combination Driver Filbeck using Route 59, the mean is 28 minutes, found by (28 + 30 + 26)/3. In a similar manner we calculate the means for the other cells and summarize the results in the following table.

	Routes			
Drivers	US 6	West End	Hickory St	Route 59
Deans	18	17	21	22
Snaverly	16	22.33	23	22
Ormson	18	23	26	22
Zollaco	23	22	29	23.67
Filbeck	25	24	28	28

The graph below shows the interaction plot using the information in the above table. The vertical axis is the travel time in minutes. The four routes are labeled on the horizontal axis and each line plots the mean travel times for each driver for all four routes. For example, the green line reports average travel times for Deans for each of the four routes.



From the graph, what observations can be made about the interaction of driver and route on travel time? Most importantly, the lines are not parallel. Because the five lines are clearly not parallel, there is an interaction effect of driver and route on travel time; that is, travel time depends on the combined effect of driver and route.

Note the differences in travel times. For the U.S. 6 route, Snaverly has the lowest or fastest mean travel time. Deans has the lowest mean travel time for the West End and Hickory Street routes. Zollaco has the slowest average time for the Hickory Street route. There are many other observations that lead to the general observation that travel time is related to the combined effects of driver AND route. The critical question is whether the observed interactions are significant or the differences are due to chance.

Hypothesis Tests for Interaction

The next step is to conduct statistical tests to further investigate the possible interaction effects. In summary, our study of travel times has several questions:

- Is there an interaction effect of routes and drivers on mean travel times?
- Are the mean travel times for drivers the same?
- Are the mean travel times for the routes the same?

Of the three questions, we are most interested in the test for interactions.

We formalize these ideas into three sets of hypotheses:

1. H_0: There is no interaction between drivers and routes.
   H_1: There is interaction between drivers and routes.
2. H_0: The driver means are equal.
   H_1: At least one driver travel time mean is different.
3. H_0: The route means are equal.
   H_1: At least one route travel time mean is different.

We test each of these hypotheses as we did in the previous section using the F distribution. The tests are summarized with the following ANOVA table. It is similar to the two-way ANOVA in the previous section with the addition of the Interaction source of variation. In addition, we refer to the driver effect as **Factor A** and the route effect as **Factor B.** Each of these hypotheses is tested using the familiar F statistic.

Source of Variation	Sum of Squares	*df*	Mean Square	*F*
Factor A (driver)	SSA	$k - 1$	$\text{MSA} = \text{SSA}/(k - 1)$	MSA/MSE
Factor B (route)	SSB	$b - 1$	$\text{MSB} = \text{SSA}/(b - 1)$	MSB/MSE
Interaction	SSI	$(k - 1)(b - 1)$	$\text{MSI} = \text{SSI}/[(k - 1)(b - 1)]$	MSI/MSE
Error	SSE	$n - kb$	$\text{MSE} = \text{SSE}/(n - kb)$	
Total		$n - 1$		

To test the hypotheses for a two-way ANOVA with interaction, we use the ANOVA: Two-Factor with Replication in the Data Analysis add-in for Excel. The details of using Excel are summarized in Appendix C. The following ANOVA table shows the results of the analysis. We use the p-values to test each hypothesis. Using the .05 significance level, the null hypotheses are rejected if the computed p-value is less than .05.

ANOVA						
Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Drivers	353.5667	4	88.39167	17.21916	0.0000	2.605975
Routes	244.9833	3	81.66111	15.90801	0.0000	2.838745
Interaction	125.7667	12	10.48056	2.041667	0.0456	2.003459
Error	205.3333	40	5.133333			
Total	929.65	59				

Reviewing the results of the ANOVA, the *p*-value for the interaction effect of .0456 is less than our significance level of .05, so our decision is to reject the null hypothesis of no interaction and conclude that the combination of route and driver has a significant effect on the response variable, travel time.

A significant interaction effect provides important information about the combined effects of the variables. If interaction is present, then a test of differences in the factor means using a one-way ANOVA for each level of the other factor is the next step. This analysis requires some time and work to complete, but the results are usually enlightening.

We will continue the analysis by conducting a one-way ANOVA for each route by testing the hypothesis H_0: Driver travel times are equal. The results follow.

US 6; H_0: Driver times are equal

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	174	4	43.5	6.04167	0.010	3.478
Within Groups	72	10	7.2			
Total	246	14				

West End; H_0: Driver times are equal

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	88.6667	4	22.1667	4.05488	0.033	3.478
Within Groups	54.6667	10	5.46667			
Total	143.333	14				

Hickory; H_0: Driver times are equal

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	135.6	4	33.9	21.1875	0.000	3.478
Within Groups	16	10	1.6			
Total	151.6	14				

Route 59; H_0: Driver times are equal

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	81.0667	4	20.2667	3.23404	0.060	3.478
Within Groups	62.6667	10	6.26667			
Total	143.733	14				

The results of the one-way ANOVA show there are significant differences in the mean travel times among the drivers for every route, except Route 59 with a *p*-value of 0.06. A review of the interaction plot may reveal some of the differences. For example, for the West End route, the graph suggests that Deans has the best mean travel time. Further statistical analysis would test pairs of mean travel times to determine the significant differences between driver travel times for each route that has a significant *p*-value.

SELF-REVIEW 12–5

See the following ANOVA table.

ANOVA					
Source of Variation	**SS**	***df***	**MS**	***F***	***p*-value**
Factor A	6.41	3	2.137	3.46	0.0322
Factor B	5.01	2	2.507	4.06	0.0304
Interaction	33.15	6	5.525	8.94	0.0000
Error	14.83	24	0.618		
Total	59.41	35			

Use the .05 significance level to answer the following questions.

(a) How many levels does Factor A have? Is there a significant difference among the Factor A means? How do you know?

(b) How many levels does Factor B have? Is there a significant difference among the Factor B means? How do you know?

(c) How many observations are there in each cell? Is there a significant interaction between Factor A and Factor B on the response variable? How do you know?

EXERCISES

19. FILE Consider the following sample data for a two-factor ANOVA analysis. There are two levels (heavy and light) of factor A (weight), and three levels (small, medium, and large) of factor B (size). For each combination of size and weight, there are three observations.

		Size		
		Small	**Medium**	**Large**
Weight	Heavy	23	20	11
		21	32	20
		25	26	20
	Light	13	20	11
		32	17	23
		17	15	8

Compute an ANOVA with statistical software, and use the .05 significance level to answer the following questions.

a. Is there a difference in the Size means?

b. Is there a difference in the Weight means?

c. Is there a significant interaction between Weight and Size?

20. Consider the following partially completed two-way ANOVA table. Suppose there are four levels of Factor A and three levels of Factor B. The number of replications per cell is 5. Complete the table and test to determine if there is a significant difference in Factor A means, Factor B means, or the interaction means. Use the .05 significance level. (Hint: estimate the values from the *F* table.)

ANOVA				
Source	**SS**	***df***	**MS**	***F***
Factor A	75			
Factor B	25			
Interaction	300			
Error	600			
Total	1000			

21. FILE A vending machine company sells its packaged foods in a variety of different machines. The company is considering three types of new vending machines. Management wants to know if the different machines affect sales. These vending machines are designated as J-1000, D-320, and UV-57. Management also wants to know if the position of the machines indoors or outdoors affects sales. Each of six similar locations was randomly assigned a machine and position combination. The data below are the number of purchases over four days.

Position/Machine	J-1000	D-320	UV-57
Inside	33, 40, 30, 31	29, 28, 33, 33	47, 39, 39, 45
Outside	43, 36, 41, 40	48, 45, 40, 44	37, 32, 36, 35

a. Draw the interaction graph. Based on your observations, is there an interaction effect? Based on the graph, describe the interaction effect of machine and position.
b. Compute an ANOVA with statistical software, and use the 0.05 level to test for position, machine, and interaction effects on sales. Report the statistical results.
c. Compare the inside and outside mean sales for each machine using statistical techniques. What do you conclude?

22. FILE A large company is organized into three functional areas: manufacturing, marketing, and research and development. The employees claim that the company pays women less than men for similar jobs. The company randomly selected four males and four females in each area and recorded their weekly salaries in dollars.

Area/Gender	Female	Male
Manufacturing	1016, 1007, 875, 968	978, 1056, 982, 748
Marketing	1045, 895, 848, 904	1154, 1091, 878, 876
Research and Development	770, 733, 844, 771	926, 1055, 1066, 1088

a. Draw the interaction graph. Based on your observations, is there an interaction effect? Based on the graph, describe the interaction effect of gender and area on salary.
b. Compute an ANOVA with statistical software, and use the 0.05 level to test for gender, area, and interaction effects on salary. Report the statistical results.
c. Compare the male and female mean salary for each area using statistical techniques. What do you recommend to the distributor?

CHAPTER SUMMARY

I. The characteristics of the F distribution are:
   A. It is continuous.
   B. Its values cannot be negative.
   C. It is positively skewed.
   D. There is a family of F distributions. Each time the degrees of freedom in either the numerator or the denominator change, a new distribution is created.

II. The F distribution is used to test whether two population variances are the same.
   A. The sampled populations must follow the normal distribution.
   B. The larger of the two sample variances is placed in the numerator, forcing the ratio to be at least 1.00.
   C. The value of F is computed using the following equation:

$$F = \frac{s_1^2}{s_2^2} \qquad \textbf{(12–1)}$$

III. A one-way ANOVA is used to compare several treatment means.

A. A treatment is a source of variation.

B. The assumptions underlying ANOVA are:

1. The samples are from populations that follow the normal distribution.

2. The populations have equal standard deviations.

3. The populations are independent.

C. The information for finding the value of F is summarized in an ANOVA table.

1. The formula for SS total, the sum of squares total, is:

$$\text{SS total} = \Sigma(x - \bar{x}_G)^2 \qquad \textbf{(12–2)}$$

2. The formula for SSE, the sum of squares error, is:

$$\text{SSE} = \Sigma(x - \bar{x}_c)^2 \qquad \textbf{(12–3)}$$

3. The formula for the SST, the sum of squares treatment, is found by subtraction.

$$\text{SST} = \text{SS total} - \text{SSE} \qquad \textbf{(12–4)}$$

4. This information is summarized in the following ANOVA table and the value of F is determined.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Treatments	SST	$k - 1$	$\text{SST}/(k - 1) = \text{MST}$	MST/MSE
Error	SSE	$n - k$	$\text{SSE}/(n - k) = \text{MSE}$	
Total	SS total	$n - 1$		

IV. If a null hypothesis of equal treatment means is rejected, we can identify the pairs of means that differ from the following confidence interval.

$$(\bar{x}_1 - \bar{x}_2) \pm t\sqrt{\text{MSE}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \qquad \textbf{(12–5)}$$

V. In a two-way ANOVA, we consider a second treatment variable.

A. The second treatment variable is called the blocking variable.

B. It is determined using the following equation:

$$\text{SSB} = k\Sigma(\bar{x}_b - \bar{x}_G)^2 \qquad \textbf{(12–6)}$$

C. The SSE term, or sum of squares error, is found from the following equation.

$$\text{SSE} = \text{SS total} - \text{SST} - \text{SSB} \qquad \textbf{(12–7)}$$

D. The F statistics for the treatment variable and the blocking variable are determined in the following table.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Treatments	SST	$k - 1$	$\text{SST}/(k - 1) = \text{MST}$	MST/MSE
Blocks	SSB	$b - 1$	$\text{SSB}/(b - 1) = \text{MSB}$	MSB/MSE
Error	SSE	$(k - 1)(b - 1)$	$\text{SSE}/[(k - 1)(b - 1)] = \text{MSE}$	
Total	SS total	$n - 1$		

VI. In a two-way ANOVA with repeated observations, we consider two treatment variables and the possible interaction between the variables. The complete ANOVA table including interactions is:

Source	Sum of Squares	df	Mean Square	F
Factor A	SSA	$k - 1$	$\text{SSA}/(k - 1) = \text{MSA}$	MSA/MSE
Factor B	SSB	$b - 1$	$\text{SSB}/(b - 1) = \text{MSB}$	MSB/MSE
Interaction	SSI	$(k - 1)(b - 1)$	$\text{SSI}/[(k - 1)(b - 1)] = \text{MSI}$	MSI/MSE
Error	SSE	$n - kb$	$\text{SSE}/(n - kb) = \text{MSE}$	
Total	SS total	$n - 1$		

PRONUNCIATION KEY

SYMBOL	MEANING	PRONUNCIATION
SS total	Sum of squares total	*S S total*
SST	Sum of squares treatment	*S S T*
SSE	Sum of squares error	*S S E*
MSE	Mean square error	*M S E*
SSB	Block sum of squares	*S S B*
SSI	Sum of squares interaction	*S S I*

CHAPTER EXERCISES

23. A real estate agent in the coastal area of Georgia wants to compare the variation in the selling price of homes on the oceanfront with those one to three blocks from the ocean. A sample of 21 oceanfront homes sold within the last year revealed the standard deviation of the selling prices was $45,600. A sample of 18 homes, also sold within the last year, that were one to three blocks from the ocean revealed that the standard deviation was $21,330. At the .01 significance level, can we conclude that there is more variation in the selling prices of the oceanfront homes?

24. One variable that Google uses to rank pages on the Internet is page speed, the time it takes for a web page to load into your browser. A source for women's clothing is redesigning their page to improve the images that show its products and to reduce its load time. The new page is clearly faster, but initial tests indicate there is more variation in the time to load. A sample of 16 different load times showed that the standard deviation of the load time was 22 hundredths of a second for the new page and 12 hundredths of a second for the current page. At the .05 significance level, can we conclude that there is more variation in the load time of the new page?

25. FILE There are two Chevrolet dealers in Jamestown, New York. The mean monthly sales at Sharkey Chevy and Dave White Chevrolet are about the same. However, Tom Sharkey, the owner of Sharkey Chevy, believes his sales are more consistent. Below are the numbers of new cars sold at Sharkey in the last 7 months and for the last 8 months at Dave White. Do you agree with Mr. Sharkey? Use the .01 significance level.

Sharkey	98	78	54	57	68	64	70	
Dave White	75	81	81	30	82	46	58	101

26. Random samples of five were selected from each of three populations. The sum of squares total was 100. The sum of squares due to the treatments was 40.
- **a.** Set up the null hypothesis and the alternate hypothesis.
- **b.** What is the decision rule? Use the .05 significance level.
- **c.** Create the ANOVA table. What is the value of *F*?
- **d.** What is your decision regarding the null hypothesis?

27. In an ANOVA table, the MSE is equal to 10. Random samples of six were selected from each of four populations, where the sum of squares total was 250.
- **a.** Set up the null hypothesis and the alternate hypothesis.
- **b.** What is the decision rule? Use the .05 significance level.
- **c.** Create the ANOVA table. What is the value of *F*?
- **d.** What is your decision regarding the null hypothesis?

28. The following is a partial ANOVA table.

Source	Sum of Squares	*df*	Mean Square	*F*
Treatment		2		
Error			20	
Total	500	11		

Complete the table and answer the following questions. Use the .05 significance level.

a. How many treatments are there?
b. What is the total sample size?
c. What is the critical value of *F*?
d. Write out the null and alternate hypotheses.
e. What is your conclusion regarding the null hypothesis?

29. FILE A consumer organization wants to know whether there is a difference in the price of a particular toy at three different types of stores. The price of the toy was checked in a sample of five discount stores, five variety stores, and five department stores. The results are shown below. Use the .05 significance level.

Discount	Variety	Department
$12	$15	$19
13	17	17
14	14	16
12	18	20
15	17	19

30. FILE Jacob Lee is a frequent traveler between Los Angeles and San Diego. For the past month, he wrote down the flight times in minutes on three different airlines. The results are:

Goust	Jet Red	Cloudtran
51	50	52
51	53	55
52	52	60
42	62	64
51	53	61
57	49	49
47	50	49
47	49	
50	58	
60	54	
54	51	
49	49	
48	49	
48	50	

a. Use the .05 significance level and the six-step hypothesis-testing process to check if there is a difference in the mean flight times among the three airlines.
b. Develop a 95% confidence interval for the difference in the means between Goust and Cloudtran.

31. FILE The City of Maumee comprises four districts. Chief of Police Andy North wants to determine whether there is a difference in the mean number of crimes committed among the four districts. He examined the records from six randomly selected days and recorded the number of crimes. At the .05 significance level, can Chief North conclude that there is a difference in the mean number of crimes among the four districts?

Number of Crimes			
Rec Center	Key Street	Monclova	Whitehouse
13	21	12	16
15	13	14	17
14	18	15	18
15	19	13	15
14	18	12	20
15	19	15	18

32. **FILE** A study of the effect of television commercials on 12-year-old children measured their attention span, in seconds. The commercials were for clothes, food, and toys. At the .05 significance level, is there a difference in the mean attention span of the children for the various commercials? Are there significant differences between pairs of means? Would you recommend dropping one of the three commercial types?

Clothes	Food	Toys
26	45	60
21	48	51
43	43	43
35	53	54
28	47	63
31	42	53
17	34	48
31	43	58
20	57	47
	47	51
	44	51
	54	

33. **FILE** When only two treatments are involved, ANOVA and the Student's *t* test (Chapter 11) result in the same conclusions. Also, for computed test statistics, $t^2 = F$. To demonstrate this relationship, use the following example. Fourteen randomly selected students enrolled in a history course were divided into two groups, one consisting of 6 students who took the course in the normal lecture format. The other group of 8 students took the course in a distance format. At the end of the course, each group was examined with a 50-item test. The following is a list of the number correct for each of the two groups.

Traditional Lecture	Distance
37	50
35	46
41	49
40	44
35	41
34	42
	45
	43

a. Using analysis of variance techniques, test H_0 that the two mean test scores are equal; $\alpha = .05$.
b. Using the *t* test from Chapter 11, compute *t*.
c. Interpret the results.

34. There are four auto body shops in Bangor, Maine, and all claim to promptly repair cars. To check if there is any difference in repair times, customers are randomly selected from each repair shop and their repair times in days are recorded. The output from a statistical software package is:

Summary				
Groups	Sample Size	Sum	Average	Variance
Body Shop A	3	15.4	5.133333	0.323333
Body Shop B	4	32	8	1.433333
Body Shop C	5	25.2	5.04	0.748
Body Shop D	4	25.9	6.475	0.595833

ANOVA					
Source of Variation	**SS**	***df***	**MS**	***F***	***p*-value**
Between Groups	23.37321	3	7.791069	9.612506	0.001632
Within Groups	9.726167	12	0.810514		
Total	33.09938	15			

Is there evidence to suggest a difference in the mean repair times at the four body shops? Use the .05 significance level.

35. The fuel efficiencies for a sample of 27 compact, midsize, and large cars are entered into a statistical software package. Analysis of variance is used to investigate if there is a difference in the mean miles per gallon of the three car sizes. What do you conclude? Use the .01 significance level.

Summary				
Groups	**Sample Size**	**Sum**	**Average**	**Variance**
Compact	12	268.3	22.35833	9.388106
Midsize	9	172.4	19.15556	7.315278
Large	6	100.5	16.75	7.303

Additional results are shown below.

ANOVA					
Source of Variation	**SS**	***df***	**MS**	***F***	***p*-value**
Between Groups	136.4803	2	68.24014	8.258752	0.001866
Within Groups	198.3064	24	8.262766		
Total	334.7867	26			

36. Three assembly lines are used to produce a certain component for an airliner. To examine the production rate, a random sample of six hourly periods is chosen for each assembly line and the number of components produced during these periods for each line is recorded. The output from a statistical software package is:

Summary				
Groups	**Sample Size**	**Sum**	**Average**	**Variance**
Line A	6	250	41.66667	0.266667
Line B	6	260	43.33333	0.666667
Line C	6	249	41.5	0.7

ANOVA					
Source of Variation	**SS**	***df***	**MS**	***F***	***p*-value**
Between Groups	12.33333	2	6.166667	11.32653	0.001005
Within Groups	8.166667	15	0.544444		
Total	20.5	17			

a. Use a .01 level of significance to test if there is a difference in the mean production of the three assembly lines.

b. Develop a 99% confidence interval for the difference in the means between Line B and Line C.

37. **FILE** The postal service sorts mail as Priority Mail Express, Priority Mail, First-Class Mail, or Standard Mail. Over a period of 3 weeks, 18 of each type were mailed from the Network Distribution Center in Atlanta, Georgia, to Des Moines, Iowa. The total delivery time in days was recorded. Minitab was used to perform the ANOVA. The results follow:

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Factor	3	35.7678	11.9226	13.62	<0.0001
Error	68	59.5167	0.8752		
Total	71	95.2844			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.935545	37.54%	34.78%	29.97%

Means

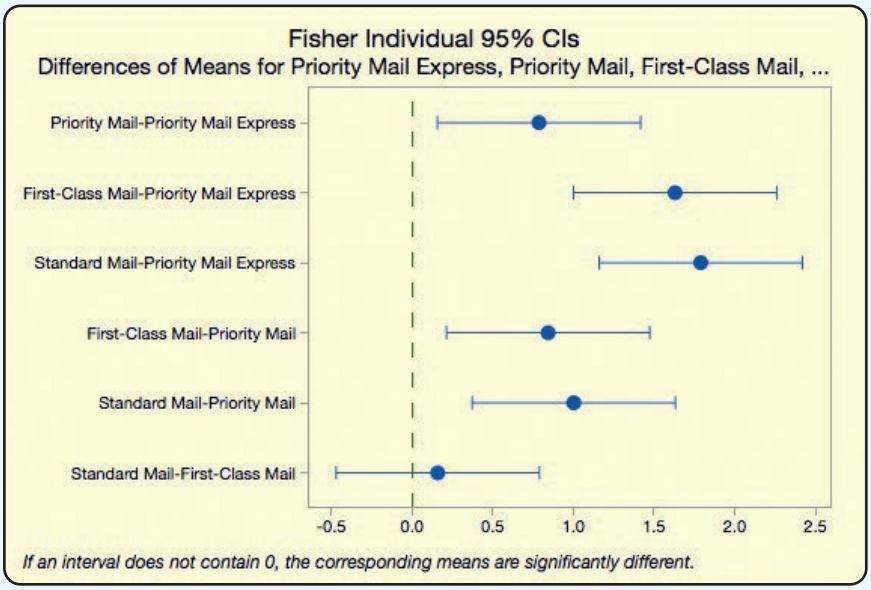
Factor	N	Mean	StDev	95% CI
Priority Mail Express	18	1.2556	0.6271	(0.8155, 1.6956)
Priority Mail	18	1.9944	0.9434	(1.5544, 2.4345)
First-Class Mail	18	2.8611	1.0600	(2.4211, 3.3011)
Standard Mail	18	3.0000	1.0460	(2.5600, 3.4400)

Pooled StDev = 0.935545

Fisher Individual Tests for Differences of Means

Difference of Levels	Difference of Means	SE of Difference	95% CI	T-Value	Adjusted P-Value
Priority Mail-Priority Mail Express	0.7389	0.3118	(0.1166, 1.3612)	2.37	0.0207
First-Class Mail-Priority Mail Express	1.6056	0.3118	(0.9833, 2.2278)	5.15	<0.0001
Standard Mail-Priority Mail Express	1.7444	0.3118	(1.1222, 2.3667)	5.59	<0.0001
First-Class Mail-Priority Mail	0.8667	0.3118	(0.2444, 1.4890)	2.78	0.0070
Standard Mail-Priority Mail	1.0056	0.3118	(0.3833, 1.6278)	3.22	0.0019
Standard Mail-First-Class Mail	0.1389	0.3118	(-0.4834, 0.7612)	0.45	0.6575

Simultaneous confidence level = 80.02%



Using the ANOVA results, compare the average delivery times of the four different types of mail.

38. **FILE** To prevent spam from entering your email inbox, you use a filter. You are interested in knowing if the number of spam emails differs by day of the week. The number of spam emails by day of week is counted and recorded. Minitab is used to perform the data analysis. Here are the results:

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Factor	6	2266.92	377.820	9.29	<0.0001
Error	48	1952.43	40.676		
Total	54	4219.35			

Means

Factor	N	Mean	StDev	95% CI
Monday	10	76.400	4.452	(72.345, 80.455)
Tuesday	9	60.444	4.333	(56.170, 64.719)
Wednesday	7	74.286	4.231	(69.439, 79.132)
Thursday	8	60.750	6.251	(56.216, 65.284)
Friday	8	74.000	7.838	(69.466, 78.534)
Saturday	5	64.000	7.036	(58.265, 69.735)
Sunday	8	69.125	9.372	(64.591, 73.659)

Pooled StDev = 6.37774

Grouping Information Using the Fisher LSD Method and 95% Confidence

Factor	N	Mean	Grouping			
Monday	10	76.400	A			
Wednesday	7	74.286	A	B		
Friday	8	74.000	A	B		
Sunday	8	69.125		B	C	
Saturday	5	64.000			C	D
Thursday	8	60.750				D
Tuesday	9	60.444				D

Means that do not share a letter are significantly different.

Fisher Individual Tests for Differences of Means

Difference of Levels	Difference of Means	SE of Difference	95% CI	T-Value	Adjusted P-Value
Tuesday-Monday	-15.956	2.930	(-21.847, -10.064)	-5.44	<0.0001
Wednesday-Monday	-2.114	3.143	(-8.434, 4.205)	-0.67	0.5044
Thursday-Monday	-15.650	3.025	(-21.733, -9.567)	-5.17	<0.0001
Friday-Monday	-2.400	3.025	(-8.483, 3.683)	-0.79	0.4315
Saturday-Monday	-12.400	3.493	(-19.424, -5.376)	-3.55	0.0009
Sunday-Monday	-7.275	3.025	(-13.358, -1.192)	-2.40	0.0201
Wednesday-Tuesday	13.841	3.214	(7.379, 20.304)	4.31	<0.0001
Thursday-Tuesday	0.306	3.099	(-5.925, 6.537)	0.10	0.9219
Friday-Tuesday	13.556	3.099	(7.325, 19.787)	4.37	<0.0001
Saturday-Tuesday	3.556	3.557	(-3.597, 10.708)	1.00	0.3226
Sunday-Tuesday	8.681	3.099	(2.450, 14.912)	2.80	0.0073
Thursday-Wednesday	-13.536	3.301	(-20.172, -6.899)	-4.10	0.0002
Friday-Wednesday	-0.286	3.301	(-6.922, 6.351)	-0.09	0.9314
Saturday-Wednesday	-10.286	3.734	(-17.794, -2.777)	-2.75	0.0083
Sunday-Wednesday	-5.161	3.301	(-11.797, 1.476)	-1.56	0.1245
Friday-Thursday	13.250	3.189	(6.838, 19.662)	4.16	0.0001
Saturday-Thursday	3.250	3.636	(-4.060, 10.560)	0.89	0.3759
Sunday-Thursday	8.375	3.189	(1.963, 14.787)	2.63	0.0115
Saturday-Friday	-10.000	3.636	(-17.310, -2.690)	-2.75	0.0084
Sunday-Friday	-4.875	3.189	(-11.287, 1.537)	-1.53	0.1329
Sunday-Saturday	5.125	3.636	(-2.185, 12.435)	1.41	0.1651

Simultaneous confidence level = 57.84%

Using the ANOVA results, compare the average number of spam emails for each day of the week.

39. **FILE** Shank's Inc., a nationwide advertising firm, wants to know whether the size of an advertisement and the color of the advertisement make a difference in the response of magazine readers. A random sample of readers is shown ads of four different colors and three different sizes. Each reader is asked to give the particular combination of size and color a rating between 1 and 10. Assume that the ratings follow the normal distribution. The rating for each combination is shown in the following table (for example, the rating for a small red ad is 2).

	Color of Ad			
Size of Ad	**Red**	**Blue**	**Orange**	**Green**
Small	2	3	3	8
Medium	3	5	6	7
Large	6	7	8	8

Is there a difference in the effectiveness of an advertisement by color and by size? Use the .05 level of significance.

40. **FILE** There are four McBurger restaurants in the Columbus, Georgia, area. The numbers of burgers sold at the respective restaurants for each of the last 6 weeks are shown below. At the .05 significance level, is there a difference in the mean number sold among the four restaurants when the factor of week is considered?

	Restaurant			
Week	**Metro**	**Interstate**	**University**	**River**
1	124	160	320	190
2	234	220	340	230
3	430	290	290	240
4	105	245	310	170
5	240	205	280	180
6	310	260	270	205

a. Is there a difference in the treatment means?
b. Is there a difference in the block means?

41. **FILE** The city of Tucson, Arizona, employs people to assess the value of homes for the purpose of calculating real estate tax. The city manager sends each assessor to the same five homes and then compares the results. The information is given below, in thousands of dollars. Can we conclude that there is a difference in the assessors? Use the .05 significance level.

	Assessor			
Home	**Zawodny**	**Norman**	**Cingle**	**Holiday**
A	$53.0	$55.0	$49.0	$45.0
B	50.0	51.0	52.0	53.0
C	48.0	52.0	47.0	53.0
D	70.0	68.0	65.0	64.0
E	84.0	89.0	92.0	86.0

a. Is there a difference in the treatment means?
b. Is there a difference in the block means?

42. **FILE** A task requires the completion of four activities. A teacher would like to know if differences in the sequence of the four activities results in different task completion times. The teacher selects three students and demonstrates the activities in random

order to the students. Then each student completes the task with each of the activity sequences. The completion times are recorded. The following table shows the minutes for each student to complete each task.

	Time (minutes)		
Sequence	Allen	Carla	Henry
A	22.4	20.8	21.5
B	17.0	19.4	20.7
C	19.2	20.2	21.2
D	20.3	18.6	20.4

Using the .05 level of significance:

a. Is there a difference in the task completion times among the different sequences?

b. Is there a difference in task completion times between the students?

43. FILE A research firm wants to compare the miles per gallon of unleaded regular, mid-grade, and super premium gasolines. Because of differences in the performance of different automobiles, seven different automobiles were selected and treated as blocks. Therefore, each brand of gasoline was tested with each type of automobile. The results of the trials, in miles per gallon, are shown in the following table. At the .05 significance level, is there a difference in the gasolines or automobiles?

Automobile	Regular	Mid-grade	Super Premium
1	21	23	26
2	23	22	25
3	24	25	27
4	24	24	26
5	26	26	30
6	26	24	27
7	28	27	32

44. FILE Each of three supermarket chains in the Denver area claims to have the lowest overall prices. As part of an investigative study on supermarket advertising, a local television station conducted a study by randomly selecting nine grocery items. Then, on the same day, an intern was sent to each of the three stores to purchase the nine items. From the receipts, the following data was recorded. At the .05 significance level, is there a difference in the mean price for the nine items between the three supermarkets?

Item	Super$	Ralph's	Lowblaws
1	$1.12	$1.02	$1.07
2	1.14	1.10	1.21
3	1.72	1.97	2.08
4	2.22	2.09	2.32
5	2.40	2.10	2.30
6	4.04	4.32	4.15
7	5.05	4.95	5.05
8	4.68	4.13	4.67
9	5.52	5.46	5.86

45. **FILE** Listed below are the weights (in grams) of a sample of M&M's Plain candies, classified according to color. Use a statistical software system to determine whether there is a difference in the mean weights of candies of different colors. Use the .05 significance level.

Red	Orange	Yellow	Brown	Tan	Green
0.946	0.902	0.929	0.896	0.845	0.935
1.107	0.943	0.960	0.888	0.909	0.903
0.913	0.916	0.938	0.906	0.873	0.865
0.904	0.910	0.933	0.941	0.902	0.822
0.926	0.903	0.932	0.838	0.956	0.871
0.926	0.901	0.899	0.892	0.959	0.905
1.006	0.919	0.907	0.905	0.916	0.905
0.914	0.901	0.906	0.824	0.822	0.852
0.922	0.930	0.930	0.908		0.965
1.052	0.883	0.952	0.833		0.898
0.903		0.939			
0.895		0.940			
		0.882			
		0.906			

46. There are four radio stations in Midland. The stations have different formats (hard rock, classical, country/western, and easy listening), but each is concerned with the number of minutes of music played per hour. From a sample of 10 randomly selected hours from each station, the sum of squared differences between each observation and the mean for its respective radio station, $\Sigma(x - \bar{x}_C)^2$, are:

Hard rock station:	126.29	Country/western station:	166.79
Classical station:	233.34	Easy listening station:	77.57

The total sum of squares for the data is: SS total = 1,099.61.

a. Determine SSE.
b. Determine SST.
c. Complete an ANOVA table.
d. At the .05 significance level, is there a difference in the treatment means?
e. If the mean for the hard rock station is 51.32 and the mean for the country/western station is 50.85, determine if there is a difference using the .05 significance level.

47. **FILE** The American Accounting Association recently conducted a study to compare the weekly wages of men and women employed in either the public or private sector of accounting. Random samples of five men and five women were selected in each group.

	Sector	
Gender	**Public**	**Private**
Men	$ 978	$1,335
	1,035	1,167
	964	1,236
	996	1,317
	1,117	1,192
Women	$ 863	$1,079
	975	1,160
	999	1,063
	1,019	1,110
	1,037	1,093

a. Draw an interaction plot of men and women means by sector.
b. Compute an ANOVA with statistical software and, using the .05 significance level, test the interaction effect of gender and sector on wages.
c. Based on your results in part (b), conduct the appropriate tests of hypotheses for differences in factor means.
d. Interpret the results in a brief report.

48. FILE Robert Altoff is vice president of engineering for a manufacturer of household washing machines. As part of a new product development project, he wishes to determine the optimal length of time for the washing cycle. Included in the project is a study of the relationship between the detergent used (four brands) and the length of the washing cycle (18, 20, 22, or 24 minutes). In order to run the experiment, 32 standard household laundry loads (having equal amounts of dirt and the same total weights) are randomly assigned to the 16 detergent–washing cycle combinations. The results (in pounds of dirt removed) are shown below.

	Cycle Time (min)			
Detergent Brand	**18**	**20**	**22**	**24**
A	0.13	0.12	0.19	0.15
	0.11	0.11	0.17	0.18
B	0.14	0.15	0.18	0.20
	0.10	0.14	0.17	0.18
C	0.16	0.15	0.18	0.19
	0.17	0.14	0.19	0.21
D	0.09	0.12	0.16	0.15
	0.13	0.13	0.16	0.17

a. Draw an interaction plot of the detergent means by cycle time.
b. Compute the ANOVA with statistical software and, using the .05 significance level, test the interaction effect of brand and cycle time on "dirt removed."
c. Based on your results in part (b), conduct the appropriate tests of hypotheses for differences in factor means.
d. Interpret the results in a brief report.

DATA ANALYTICS

49. FILE The North Valley Real Estate data reports information on the homes sold last year.
a. At the .02 significance level, is there a difference in the variability of the selling prices of the homes that have a pool versus those that do not have a pool?
b. At the .02 significance level, is there a difference in the variability of the selling prices of the homes with an attached garage versus those that do not have an attached garage?
c. At the .05 significance level, is there a difference in the mean selling price of the homes among the five townships?
d. Adam Marty recently joined North Valley Real Estate and was assigned twenty homes to market and show. When he was hired, North Valley assured him that the twenty homes would be fairly assigned to him. When he reviewed the selling prices of his assigned homes, he thought that the prices were much below the average of $357,000. Adam was able to find the data of the homes assigned to agents in the firm. Use statistical inference to compare the mean price of homes assigned to him to the mean price of homes assigned to the other agents. What do the results indicate? How is your analysis defining fairness?
e. Home buyers finance the purchase of their home with a mortgage. In this data, the mortgages are either a fixed rate mortgage paid over 30 years, or an adjustable rate mortgage. The adjustable rate mortgage provides a lower introductory interest rate for the first five years of occupancy. Then, in the fifth year, the rate is adjusted to the current rate plus an additional percent. Usually, the adjusted rate is higher than

the "introductory" rate. With this information, we may predict that the average years of occupancy would be different for home owners based on the type of mortgage and whether they defaulted on the mortgage. Use the data to evaluate this prediction.

50. FILE Refer to the Baseball 2016 data, which report information on the 30 Major League Baseball teams for the 2016 season.

a. At the .10 significance level, is there a difference in the variation in team salary among the American and National League teams?

b. Create a variable that classifies a team's total attendance into three groups: less than 2.0 (million), 2.0 up to 3.0, and 3.0 or more. At the .05 significance level, is there a difference in the mean number of games won among the three groups?

c. Using the same attendance variable developed in part (b), is there a difference in the mean number of home runs hit per team? Use the .05 significance level.

d. Using the same attendance variable developed in part (b), is there a difference in the mean salary of the three groups? Use the .05 significance level.

51. FILE Refer to the Lincolnville School District bus data.

a. Conduct a test of hypothesis to reveal whether the mean maintenance cost is equal for each of the bus manufacturers. Use the .01 significance level.

b. Conduct a test of hypothesis to determine whether the mean miles traveled since the last maintenance is equal for each bus manufacturer. Use the .05 significance level.

A REVIEW OF CHAPTERS 10–12

This section is a review of the major concepts and terms introduced in Chapters 10, 11, and 12. Chapter 10 began our study of hypothesis testing. A hypothesis is an assumption about a characteristic of a population. In statistical hypothesis testing, we begin by making a statement about the value of the population parameter in the null hypothesis. We establish the null hypothesis for the purpose of testing. When we complete the testing, our decision is either to reject or to fail to reject the null hypothesis. If we reject the null hypothesis, we conclude that the alternate hypothesis is true. The alternate hypothesis is "accepted" only if we show that the null hypothesis is false. We also refer to the alternate hypothesis as the research hypothesis. Most of the time we want to prove the alternate hypothesis.

In Chapter 10, we selected random samples from a single population and tested whether it was reasonable that the population parameter under study equaled a particular value. For example, we wish to investigate whether the mean tenure of those holding the position of CEO in large firms is 12 years. We select a sample of CEOs, compute the sample mean, and compare the mean of the sample to the population. The single population under consideration is the length of tenure of CEOs of large firms. We described methods for conducting the test when the population standard deviation was available and when it was not available.

In Chapter 11, we extended the idea of hypothesis testing to whether two independent random samples came from populations having the same or equal population means. For example, St. Mathews Hospital operates an urgent care facility on both the north and south sides of Knoxville, Tennessee. The research question is: Is the mean waiting time for patients visiting the two facilities the same? To investigate, we select a random sample from each of the facilities and compute the sample means. We test the null hypothesis that the mean waiting time is the same at the two facilities. The alternate hypothesis is that the mean waiting time is not the same for the two facilities. If the population standard deviations are known, we use the z distribution as the test statistic. If the population standard deviations are not known, the test statistic follows the t distribution.

Our discussion in Chapter 11 also concerned dependent samples. The test statistic is the t distribution and we assume that the distribution of differences follows the normal distribution. One typical paired sample problem calls for recording an individual's blood pressure before administering medication and then again afterward in order to evaluate the effectiveness of the medication. We also considered the case of testing two population proportions. For example, the production manager wished to compare the proportion of defects on the day shift with that of the second shift.

Chapter 11 dealt with the difference between two population means. Chapter 12 presented tests for variances and a procedure called the *analysis of variance,* or *ANOVA*. ANOVA is used to simultaneously determine whether several independent normal populations have the same mean. This is accomplished by comparing the variances of the random samples selected from these populations. We apply the usual hypothesis-testing procedure, but we use the F distribution as the test statistic. Often the calculations are tedious, so a statistical software package is recommended.

As an example of analysis of variance, a test could be conducted to resolve whether there is any difference in effectiveness among five fertilizers on the weight of popcorn ears. This type of analysis is referred to as *one-factor ANOVA* because we are able to draw conclusions about only one factor, called a *treatment*. If we want to draw conclusions about the simultaneous effects of more than one factor or variable, we use the *two-factor ANOVA* technique. Both the one-factor and two-factor tests use the *F distribution* as the distribution of the test statistic. The *F* distribution is also the distribution of the test statistic used to find whether one normal population has more variation than another.

An additional feature of the two-factor ANOVA is the possibility that interactions may exist between the factors. There is an *interaction* if the response to one of the factors depends on the level of the other factor. Fortunately, the ANOVA is easily extended to include a test for interactions.

PROBLEMS

For problems 1–6, state: (a) the null and the alternate hypotheses, (b) the decision rule, and (c) the decision regarding the null hypothesis, (d) then interpret the result.

1. A machine is set to produce tennis balls so the mean bounce is 36 inches when the ball is dropped from a platform of a certain height. The production supervisor suspects that the mean bounce has changed and is less than 36 inches. As an experiment, a sample of 12 balls was dropped from the platform and the mean height of the bounce was 35.5 inches, with a standard deviation of 0.9 inch. At the .05 significance level, can the supervisor conclude that the mean bounce height is less than 36 inches?
2. FILE It was hypothesized that road construction workers, on the average, spend 20 minutes of each hour not engaged in productive work. Some claimed the nonproductive time is greater than 20 minutes. An actual study was conducted at a construction site, using a stopwatch and other ways of checking the work habits. A random check of workers revealed the following unproductive times, in minutes, during a one-hour period (exclusive of regularly scheduled breaks):

10	25	17	20	28	30	18	23	18

   Using the .05 significance level, is it reasonable to conclude the mean unproductive time is greater than 20 minutes?
3. Stiktite, Inc. plans a test of the mean holding power of two glues designed for plastic. First, a small plastic hook was coated at one end with Epox glue and fastened to a sheet of plastic. After it dried, weight was added to the hook until it separated from the sheet of plastic. The weight was then recorded. This was repeated until 12 hooks were tested. The same procedure was followed for Holdtite glue, but only 10 hooks were used. The sample results, in pounds, were:

	Epox	Holdtite
Sample size	12	10
Sample mean	250	252
Sample standard deviation	5	8

   At the .01 significance level, is there a difference between the mean holding power of Epox and that of Holdtite?
4. FILE Pittsburgh Paints wishes to test an additive formulated to increase the life of paints used in the hot and arid conditions of the Southwest. The top half of a piece of wood was painted using the regular paint. The bottom half was painted with the paint including the additive. The same procedure was followed for a total of 10 pieces. Then each

piece was subjected to brilliant light. The data, the number of hours each piece lasted before it faded beyond a certain point, follow:

	Number of Hours by Sample									
	A	B	C	D	E	F	G	H	I	J
Without additive	325	313	320	340	318	312	319	330	333	319
With additive	323	313	326	343	310	320	313	340	330	315

Using the .05 significance level, determine whether the additive is effective in prolonging the life of the paint.

5. FILE A Buffalo, New York, cola distributor is featuring a special sale on 12-packs. She wonders where in the grocery store to place the cola for maximum attention. Should it be near the front door of the grocery stores, in the cola section, at the checkout registers, or near the milk and other dairy products? Four stores with similar total sales cooperated in an experiment. In one store, the 12-packs were stacked near the front door, in another they were placed near the checkout registers, and so on. Sales were checked at specified times in each store for exactly four minutes. The results were:

Cola at the Door	In Soft Drink Section	Near Registers	Dairy Section
$6	$ 5	$ 7	$10
8	10	10	9
3	12	9	6
7	4	4	11
	9	5	
		7	

The Buffalo distributor wants to find out whether there is a difference in the mean sales for cola stacked at the four locations in the store. Use the .05 significance level.

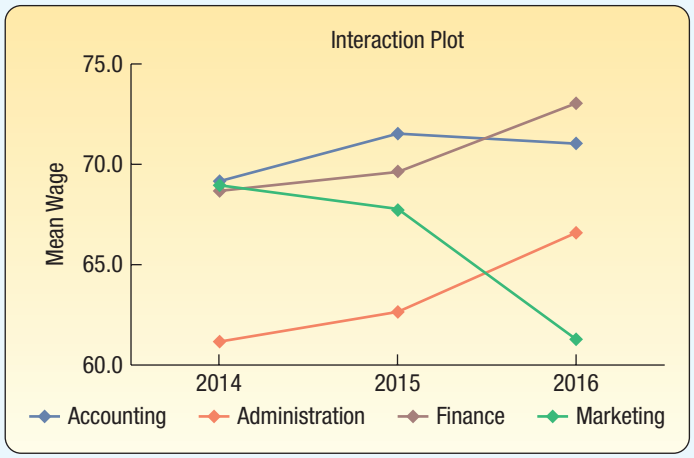
6. FILE Williams Corporation is investigating the effects of educational background on employee performance. A potential relevant variable in this case is the self-rated social status of the employee. The company has recorded the annual sales volumes (in $000) achieved by sales employees in each of the categories below. Perform a complete two-way analysis of variance (including the possibility of interactions) on the data and describe what your results suggest.

	School Type		
Self-Rated Social Status	Ivy League	State-supported	Small Private
Low	62, 61	68, 64	70, 70
Medium	68, 64	74, 68	62, 65
High	70, 71	57, 60	57, 56

7. A school supervisor is reviewing initial wages of former students (in $000). Samples were taken over 3 years for four different majors (accounting, administration, finance, and marketing). For each combination of major and year, three former students were sampled.

Major/Year	2014	2015	2016
Accounting	75.4, 69.8, 62.3	73.9, 78.8, 62.0	64.2, 80.8, 68.2
Administration	61.5, 59.9, 62.1	63.9, 57.6, 66.5	74.2, 67.5, 58.1
Finance	63.6, 70.2, 72.2	69.2, 72.5, 67.2	74.7, 66.4, 77.9
Marketing	71.3, 69.2, 66.4	74.0, 67.6, 61.7	60.0, 61.3, 62.5

a. Here is an interaction plot of the information. What does it reveal?



b. Write out all of the pairs of null and alternative hypotheses you would apply for a two-way ANOVA.

c. Here is the statistical software output. Use the 0.05 level to check for interactions.

Source	DF	SS	MS	F	P
Major	3	329.20	109.732	3.39	0.034
Year	2	7.32	3.659	0.11	0.894
Interaction	6	183.57	30.595	0.94	0.482
Error	24	777.29	32.387		
Total	35	1297.37			

d. If proper, test the other hypotheses at the .05 significance level. If it is not appropriate, describe why you should not do the tests.

CASES

A. Century National Bank

Refer to the description of Century National Bank on page 129 at the end of the Review of Chapters 1–4.

With many other options available, customers no longer let their money sit in a checking account. For many years the mean checking balance has been $1,600. Do the sample data indicate that the mean account balance has declined from this value?

Recent years have also seen an increase in the use of ATM machines. When Mr. Selig took over the bank, the mean number of transactions per month per customer was 8; now he believes it has increased to more than 10. In fact, the advertising agency that prepares TV commercials for Century would like to use this on the new commercial being designed. Is there sufficient evidence to conclude that the mean number of transactions per customer is more than 10 per month? Could the advertising agency say the mean is more than 9 per month?

The bank has branch offices in four different cities: Cincinnati, Ohio; Atlanta, Georgia; Louisville, Kentucky; and Erie, Pennsylvania. Mr. Selig would like to know whether there is a difference in the mean checking account balances among the four branches. If there are differences, between which branches do these differences occur?

Mr. Selig is also interested in the bank's ATMs. Is there a difference in ATM use among the branches? Also, do customers who have debit cards tend to use ATMs differently from those who do not have debit cards? Is there a difference in ATM use by those with checking accounts that pay interest versus those that do not? Prepare a report for Mr. Selig answering these questions.

B. Bell Grove Medical Center

Ms. Gene Dempsey manages the emergency care center at Bell Grove Medical Center. One of her responsibilities is to have enough nurses so that incoming patients needing service can be handled promptly. It is stressful for patients to wait a long time for emergency care even when their care needs are not life threatening. Ms. Dempsey gathered the following information regarding the number of patients over the last several weeks. The center is not open on weekends. Does it appear that there are any differences in the number of patients served by the day of the week? If there are differences, which days seem to be the busiest? Write a brief report summarizing your findings.

Date	Day	Patients
9-29-16	Monday	38
9-30-16	Tuesday	28
10-1-16	Wednesday	28
10-2-16	Thursday	30
10-3-16	Friday	35
10-6-16	Monday	35
10-7-16	Tuesday	25
10-8-16	Wednesday	22
10-9-16	Thursday	21
10-10-16	Friday	32
10-13-16	Monday	37
10-14-16	Tuesday	29
10-15-16	Wednesday	27
10-16-16	Thursday	28
10-17-16	Friday	35
10-20-16	Monday	37
10-21-16	Tuesday	26
10-22-16	Wednesday	28
10-23-16	Thursday	23
10-24-16	Friday	33

PRACTICE TEST

Part 1—Objective

1. A statement about the value of a population parameter that always includes the equal sign is called the _____. 1. _____
2. The likelihood of rejecting a true null hypothesis is called the _____. 2. _____
3. Assuming the null hypothesis is true, the likelihood of finding a value of the test statistic at least as extreme as the one found in the sample is called the _____. 3. _____
4. When conducting a test of hypothesis about a single population mean, the z distribution is used as the test statistic only when the _____ is known. 4. _____
5. In a two-sample test of hypothesis for means, if the population standard deviations are not known, *what must we assume about the shape of the populations?* 5. _____
6. A value calculated from sample information used to determine whether to reject the null hypothesis is called the _____. 6. _____
7. In a two-tailed test, the rejection region is _____. (all in the upper tail, all in the lower tail, split evenly between the two tails, none of these—pick one) 7. _____
8. Which of the following is not a characteristic of the F distribution? (continuous, positively skewed, range from $-\infty$ to ∞, family of distributions) 8. _____
9. To perform a one-way ANOVA, the treatments must be _____. (independent, mutually exclusive, continuous) 9. _____
10. In a one-way ANOVA, there are four treatments and six observations in each treatment. What are the degrees of freedom for the F distribution? 10. _____

Part 2—Problems

For problems 1 and 2, state the null and alternate hypotheses and the decision rule, make a decision regarding the null hypothesis, and interpret the result.

1. The Park Manager at Fort Fisher State Park in North Carolina believes the typical summer visitor spends more than 90 minutes in the park. A sample of 18 visitors during the months of June, July, and August revealed the mean time in the park for visitors was 96 minutes, with a standard deviation of 12 minutes. At the .01 significance level, is it reasonable to conclude the mean time in the park is greater than 90 minutes?
2. Is there a difference in the mean miles traveled per week by each of two taxicab companies operating in the Grand Strand area? The *Sun News,* the local paper, is investigating and obtained the following sample information. At the

.05 significance level, is it reasonable to conclude there is a difference in the mean miles traveled? Assume equal population variances.

Variable	Yellow Cab	Horse and Buggy Cab Company
Sample size	14	12
Mean miles	837	797
Standard deviation	30	40

3. The results of a one-way ANOVA are reported below. Use the .05 significance level.

ANOVA				
Source of Variation	**SS**	***df***	**MS**	***F***
Between groups	6.892202	2	3.446101	4.960047
Within groups	12.50589	18	0.694772	
Total	19.3981	20		

Answer the following questions.

a. How many treatments are in the study?
b. What is the total sample size?
c. What is the critical value of *F*?
d. Write out the null hypothesis and the alternate hypothesis.
e. What is your decision regarding the null hypothesis?
f. Why can we conclude the treatment means differ?